The Finite-Harmonic Formal Second-Order Tangent Cone of Compact Weyl–Maxwell Gravity

Moment Maps, Bounded Resonances, and a Balanced Cubic Test

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Abstract

Linearized solutions of a nonlinear gauge theory need not be tangent to exact solutions. We classify this failure at second order for finite harmonic configurations of Weyl–Maxwell gravity on the compact magnetically supported product $\mathbb{R}_t \times S^1 \times S^2$. The certified linear inventory contains the ordinary Einstein–Maxwell branches, two additional fourth-order branches in each generic parity, exceptional dipoles, a homogeneous generalized block, and axial twist modes.

For spatially periodic corrections with finite exponential-polynomial time dependence, the complete second-order obstruction space is five-dimensional. Its covectors are precisely the Taub pairings with time translation, circle translation, and the three sphere rotations. Thus the complete finite-support exponential-polynomial tangent cone is

$$\mathscr{Z}_2^{\mathrm{ep}} = \{u : \mu_H(u) = \mu_{P_x}(u) = \mu_{J_1}(u) = \mu_{J_2}(u) = \mu_{J_3}(u) = 0\}.$$

This is a necessary-and-sufficient theorem in the declared correction class, not merely a list of examples.

Bounded corrections obey additional conditions. Polynomial-growth source coefficients and nonzero-shell adjoint pairings survive independently of the five moment maps. We classify the complete standard generalized-zero subcone, prove that electric-charge and flat-Wilson oscillator columns are bounded-removable, and determine the full circumference column: a nonzero circle-radius tangent is compatible with a finite oscillator precisely when that oscillator has zero circle momentum. A repaired full-time circumference-velocity fixture also exposes a nonzero polar polynomial obstruction that was invisible at $t = 0$; its full a, d-cross ideal on the $\ell = 2, k = 0$ extra block is now classified. On the complete global-plus-$\ell = 2, k = 0$-extra carrier, the bounded cone then collapses exactly to the standard static-global cone: no nonzero extra amplitude survives. At one tuned nonzero momentum, by contrast, we classify the complete axisymmetric all-primary bounded cone. Its only shell resonance cuts out two mixed axial–polar Einstein-minus sheets, and the lower-frequency extra primary widens the exact interval on which positive-primary occupations can balance their energy and momentum. The remaining oscillator columns are stated as open gates. A successor theorem promotes the standard-branch part of this calculation to every $\ell \geq 2$: the exact two-parity resonance matrix has two mixed sheets and two one-sided planes, and charge balance reduces the complete tuned axisymmetric bounded cone to the origin together with the two mixed sheets on an explicit occupation interval. A separate two-absolute-momentum computation has certified all 108 axisymmetric $L = 4$ branch-basis coefficients and all 56 declared nonaxisymmetric $L = 1, 3$ coefficients; every individual basis

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fixture is obstructed, while amplitude-level cancellations remain open. Consequently formal exponential-polynomial extendibility is a moment-map theorem, whereas bounded quasiperiodic extendibility is a strict resonance problem.

For one balanced real Einstein-minus/additional-primary tangent, we also evaluate the first action-normalized third-order obstruction. Its intrinsic compact-stabilizer class vanishes in the correction-independent quotient $\mathfrak{o}/\operatorname{im}\ell_2$, but the certified second-order representative has nonzero adjoint pairings on every occupied original characteristic shell. Hence it has no bounded finite-quasiperiodic third-order correction, while a finite exponential-polynomial correction with controlled secular degree exists. The shell verdict is scoped to that second-order representative; its quotient by arbitrary homogeneous second-order additions remains open.

All results are LOCAL-ALGEBRAIC and REDUCED-MODE. They do not establish an all-orders family, a causal retarded correction, a final residual quotient, or a quantum state space.

AI authorship and computational accountability

The named AI systems developed the programme, derivations, symbolic implementations, independent verification architecture, claim audits, and this manuscript. Asger Alstrup Palm commissioned and coordinated the work and is the corresponding human contact. Every theorem stated as certified below is bound to an exact machine-readable certificate and a separately implemented verifier. Exact rational or algebraic arithmetic is used for all ranks, cokernels, polynomial reductions, and source remainders. The manuscript remains conditional on final human review of its mathematical scope, references, and exposition.

Contents

1 Question, answer, and claim boundary

Let $\Phi(s)$ be a putative curve of solutions of the Weyl–Maxwell equations $\mathcal{E}(\Phi) = 0$, expanded as

$$\Phi(s) = \bar{\Phi} + su + s^2 v + O(s^3).$$

The first two equations are

$$\mathcal{L}u = 0, \qquad \mathcal{L}v = -\mathcal{E}^{(2)}_{\bar{\Phi}}(u,u), \qquad \mathcal{L} := D\mathcal{E}_{\bar{\Phi}}, \qquad \mathcal{E}^{(2)}_{\bar{\Phi}} := \frac{1}{2}D^2\mathcal{E}_{\bar{\Phi}}. \tag{1}$$

The first equation describes the linear solution space. The second asks whether the quadratic source lies in the image of the linearized operator in a specified function class. Changing that function class changes the answer: a resonant source may have a finite exponential-polynomial secular primitive but no bounded quasiperiodic primitive.

The paper answers the following finite-harmonic question.

> For the complete certified linear inventory on the compact product, which real finite-support tangents admit a spatially periodic second-order correction, and which additional conditions appear when that correction is required to remain bounded in time?

The answer has two layers:

$$u \in \mathcal{Z}_2^{\mathrm{ep}} \iff \mu_H = \mu_{P_x} = \mu_{J_1} = \mu_{J_2} = \mu_{J_3} = 0, \tag{2}$$

$$u \in \mathcal{Z}_2^{\mathrm{qp}} \iff u \in \mathcal{Z}_2^{\mathrm{ep}}, \quad \mathcal{P}_{j,r}(u) = 0, \quad \mathcal{R}_{j,a}(u) = 0. \tag{3}$$

Here $\mathcal{P}_{j,r}$ are positive polynomial-growth coefficients and $\mathcal{R}_{j,a}$ are reduced adjoint pairings on nonzero characteristic shells. Both obstruction ledgers are complete in the declared finite carrier. The paper gives several complete columns of the second line but does not claim a classification of the common quadratic zero locus of all bounded obstructions.

At third order we make one further, fixture-scoped statement. For the balanced tangent u and its certified no-homogeneous-addition correction v,

$$[K_3(u;v)] = 0 \quad \text{in} \quad \mathfrak{o}/\operatorname{im}\ell_2(u,-), \qquad \mathcal{R}^{(3)}_{\mathrm{occupied}}(u;v) \neq 0. \tag{4}$$

The first statement is correction independent. The second proves bounded third-order obstruction for the displayed representative only; finite exponential-polynomial inversion with secular terms remains possible.

1.1 Dependency typing

The spacetime is Lorentzian, but the proof is an exact harmonic and finite-module calculation. It carries the dependency tags

LOCAL-ALGEBRAIC REDUCED-MODE.

No theorem below carries LORENTZIAN-CAUSAL. In particular, a finite exponential-polynomial right inverse is not a retarded Green operator, and a bounded quasiperiodic correction is not an all-orders nonlinear solution.

2 Relation to classical linearization stability

The structural origin of the five quadratic conditions is classical. Brill and Deser exhibited nonintegrable linearized solutions on closed spatial sections [1]; Fischer and Marsden formulated linearization stability as a nonlinear functional-analytic problem [2]; and Moncrief identified spacetime symmetries with the adjoint-kernel constraints governing second-order integrability [3, 4]. Arms treated the coupled Einstein–Maxwell problem on the appropriate electromagnetic bundle [5]. The Arms–Marsden–Moncrief momentum-map normal form then described the local solution space near a symmetric point as a smooth factor times the zero set of a homogeneous quadratic map [6, 8]. Higher-curvature Taub constraints and their second-order interpretation have also been developed in general form [10, 11].

Accordingly, this paper does *not* claim to discover the general principle "tangent cone equals moment-map zero." Its contribution is an explicit exhaustive realization in the declared fourth-order Weyl–Maxwell model: the complete certified finite carrier, a proof that exactly five covectors exhaust the exponential-polynomial cokernel, and a bounded refinement by independent polynomial-growth and characteristic-shell obstructions. The latter belongs to the broader problem of secular perturbation theory [12], but the concrete $\{\mu, \mathcal{P}, \mathcal{R}\}$ ledger and exceptional mixed resonance are model-specific results.

The third-order contribution is narrower and logically separate: for one balanced mixed-primary fixture it computes the action-normalized cubic compact-stabilizer quotient and the original-shell Fredholm pairings in the same calculation. The resulting contrast—zero intrinsic global class but nonzero bounded shell obstruction for the certified second-order representative—is not supplied by the general momentum-map normal form or by the existence of a secular right inverse. To our knowledge, this exact branch-resolved comparison has not previously been exhibited for the compact Weyl–Maxwell product.

The relation is conceptual rather than a claim that the classical functional-analytic normal-form theorem has been reproved for this theory. A separate action-derived compact-Cauchy audit, using the canonical Weyl variables of Refs. [14, 15], finds that the raw seven-row Weyl–Maxwell constraint symbol is underdetermined elliptic on the fixed bundle. It therefore has closed range and finite-dimensional cokernel on the compact slice. After seven legitimate local gauge conditions are appended, however, the symbol still maps a rank 30 phase-space bundle to rank 14 and has an exact rank 16 physical kernel, including two transverse-traceless Ostrogradsky-momentum directions. Thus this constraint-plus-gauge operator is right semi-Fredholm, not Fredholm. The global adjoint kernel can nevertheless be determined exactly. For the canonical constraint generator, the Hamiltonian transpose identity identifies $\ker(D\mathcal{C}_{\bar{z}})^*$ with the infinitesimal stabilizer algebra of the background canonical data. The complete Fourier–spherical decomposition leaves only H, P_x in the $(k, \ell) = (0, 0)$ block and the three coexact $\ell = 1$ rotations in the $k = 0$ block. Each rotation includes its unique based bundle lift $\iota_J \bar{F} + d\chi_J = 0$. Hence

$$\ker(D\mathcal{C}_{\bar{z}})^* = \operatorname{span}_{\mathbb{R}}\{H, P_x, J_1, J_2, J_3\}.$$

The constant Maxwell reducibility is removed by the mean-zero Gauss codomain and is not a sixth Taub covector. This closes the compact-Cauchy adjoint classification. Closed range then gives orthogonal Hilbert splittings of the domain and target, so a Banach Kuranishi reduction applies. Its quadratic obstruction is precisely the projection onto these five adjoint covectors. Consequently the full Sobolev second-order *Cauchy* tangent cone is the common zero of the five Taub maps, with their vanishing necessary and sufficient for a Cauchy second-order correction.

There is nevertheless a sharp boundary to the usual momentum-map slogan. On the unextended canonical phase space the normal-deformation bracket contains the structure function

$$\{H[N], H[M]\} = H_i\big[h^{ij}(N\partial_j M - M\partial_j N)\big] + \cdots .$$

Thus the full lapse sector is not the infinitesimal action of a fixed Lie group on a phase-space neighborhood, one of the hypotheses of the direct Arms–Marsden–Moncrief momentum-map normal form. For example, $N = \sin x$, $M = \cos x$ gives shift coefficient -1 at $h_{xx} = 1$ and $-1/4$ at $h_{xx} = 4$. The Kuranishi obstruction may therefore have higher terms: identifying the exact nonlinear zero set with the homogeneous quadratic cone requires embedding variables, an algebroid/groupoid slice, or a Weyl–Maxwell-specific extension of the several-Killing-field theorem.

The classical normal-form theorems use Sobolev-completed initial-data manifolds, a split constraint map, gauge slices, and control of the full adjoint kernel. The compact audit now supplies the split Kuranishi reduction and its complete second-order cone, but not an exact homogeneous-quadratic all-orders normal form. Here we instead prove an algebraic surjectivity and cokernel theorem on a complete *declared finite harmonic module* of the fourth-order Weyl–Maxwell equations. We do not establish the hypotheses needed to promote it to an infinite-mode Sobolev normal form. Conversely, the classical structural theorem does not by itself supply our block inventory, five-covector sufficiency statement, or bounded $\mathcal{P}/\mathcal{R}$ refinement.

3 Theory and common background

We use the compactified Plebański–Hacyan product

$$M = \mathbb{R}_t \times S^1_L \times S^2, \tag{5}$$

with unit sphere curvature and a parallel unit magnetic flux on a fixed compact $U(1)$ bundle. With the conventions of the companion linear phase-space paper, the background solves both Einstein–Maxwell and Weyl–Maxwell theory. The latter has local gauge group

$$\mathrm{Diff}(M) \times \mathrm{Weyl}(M) \times U(1).$$

The spatial slice $\Sigma = S^1 \times S^2$ is compact. Its background stabilizer algebra contains

$$H = \partial_t, \qquad P_x = \partial_x, \qquad J_1, J_2, J_3 \in \mathfrak{so}(3).$$

For a linear tangent u, the corresponding covariant phase-space moment maps are quadratic forms

$$\mu_X(u) = \frac{1}{2}\Omega(u, \mathcal{L}_X u), \qquad X \in \{H, P_x, J_1, J_2, J_3\}.$$

On a closed slice, the covariant Taub bridge identifies these with the pairings of the quadratic field-equation source against the stabilizer cokernel covectors.

Let $\mathcal{C}$ denote the spatial constraint projection of the Euler–Lagrange operator and put $\mathcal{C}^{(2)}_{\bar\Phi} := \frac{1}{2}D^2\mathcal{C}_{\bar\Phi}$. For a constraint source S_A, our pairing convention is $\langle \zeta, S\rangle_\Sigma = \int_\Sigma \zeta^A S_A \, \mathrm{d}\Sigma$.

Proposition 3.1 (Taub moment maps on ordinary and generalized modes)**.** *The five displayed automorphisms lift to the fixed magnetic bundle and act symplectically on every certified linear block. For every linear solution, including the homogeneous and twist Jordan blocks,*

$$\left\langle \zeta_X, \mathcal{C}^{(2)}_{\bar\Phi}(u,u)\right\rangle_\Sigma = \mu_X(u) = \frac{1}{2}\Omega_\Sigma(u, \mathcal{L}_X u), \qquad X \in \{H, P_x, J_1, J_2, J_3\}. \tag{6}$$

The right side is independent of the closed Cauchy slice Σ. There is no sixth $U(1)$ Taub covector in the fixed-bundle problem.

Proof. Twice differentiating the action Noether identity pairs the quadratic constraint source with the adjoint reducibility parameter and gives (6). Because u and $\mathcal{L}_X u$ solve the linearized equations, their Lee–Wald current is conserved. The difference between its integrals on two closed Cauchy slices is therefore the integral of an exact spacetime divergence and vanishes. This argument uses neither a frequency eigenbasis nor boundedness, so it applies to polynomial Jordan solutions.

Concretely, time translation acts on the homogeneous coordinates by

$$(a, b, c, d, Q_e, W_x) \longmapsto (b, 0, d, 2a, 0, Q_e)$$

and on each twist pair by $(A, B) \mapsto (B, 0)$. Substitution into the conserved Cauchy form gives, after omission of the common positive harmonic normalizations,

$$\mu_H^{\ell=0} = -a^2 - b^2 + bd - Q_e^2, \tag{7}$$

$$\mu_H^{\text{twist}} = 2|B|^2, \qquad \mu_{\mathbf{J}}^{\text{twist}} = -4A \times B. \tag{8}$$

Although the representatives contain powers of t, these expressions are constant; the apparent time-dependent contributions cancel in the conserved current.

The rotations lift as bundle automorphisms $\widehat{J}_a = (J_a, \chi_a)$, with $\iota_{J_a}\bar F + \mathrm{d}\chi_a = 0$ patchwise. A constant $U(1)$ parameter acts trivially on connection differences and is a reducibility parameter, not an independent Hamiltonian stabilizer. Finally, fixing the Chern class of the magnetic bundle forbids a harmonic magnetic variation. These facts leave exactly the five covectors in (6). □

4 The complete finite harmonic carrier

The real finite-support carrier V_{fin} is the direct sum of the following certified linear blocks:

1. generic Einstein and extra primaries for every $\ell \geq 2$, parity, magnetic label, and compact momentum;
2. standard and extra exceptional dipoles for every compact momentum;
3. the axial twist Jordan block;
4. the six-coordinate homogeneous generalized block; and
5. the electric and Wilson global directions on the fixed bundle.

Reality pairs opposite frequencies and momenta. A finite input therefore produces a finite quadratic source. Its complete output decomposition is indexed by

$$(L, M, K, \Omega, \text{parity}).$$

Angular Clebsch–Gordan selection gives finitely many L, while temporal and circle products give finite sums of input frequencies and momenta.

Definition 4.1 (Finite correction modules)**.** Let $\mathscr{X}_{\text{ep}}$ be the spatially periodic fields whose time coefficients are finite sums of $t^p e^{-i\Omega t}$, with $p \in \mathbb{N}_0$ and a finite set of real frequencies. Let $\mathscr{X}_{\text{qp}} \subset \mathscr{X}_{\text{ep}}$ be the finite bounded quasiperiodic subclass with $p = 0$. Define

$$\mathcal{Z}_2^{\mathscr{X}} = \{u \in V_{\text{fin}} : \exists v \in \mathscr{X},\ \mathcal{L}v = -\mathcal{E}^{(2)}(u,u)\}.$$

Every element of $\mathscr{X}_{\text{ep}}$ is smooth, but $\mathscr{X}_{\text{ep}}$ is not the full C^∞, Sobolev, distributional, or retarded function space. The subscripts "ep" and "qp" are therefore load-bearing. A secular coefficient is finite at every finite time but prevents a uniformly bounded perturbative expansion. It may signal a nearby family with shifted frequency, a required renormalization of parameters, or genuine resonant growth. The certificates decide only these two finite-module dispositions.

5 Finite exponential-polynomial tangent-cone theorem

The sufficiency statement rests on a simple algebraic fact which we state explicitly because it is the step that distinguishes $\mathscr{X}_{\text{ep}}$ from $\mathscr{X}_{\text{qp}}$.

Lemma 5.1 (Exponential-polynomial surjectivity)**.** *Let $P(D_t) \neq 0$ be a constant-coefficient scalar differential operator. For every polynomial $f(t)$ and real Ω, there is a polynomial $g(t)$ such that*

$$P(D_t)\big(g(t)e^{-i\Omega t}\big) = f(t)e^{-i\Omega t}. \tag{9}$$

If $-i\Omega$ is a root of P of multiplicity m, one may take $\deg g \leq \deg f + m$. Consequently every nonzero invariant factor of a matrix operator over the fibre ring $K[D_t]$ is surjective on $\mathscr{X}_{\text{ep}}$; only zero Smith factors contribute to its $\mathscr{X}_{\text{ep}}$-cokernel.

Proof. Conjugation by $e^{-i\Omega t}$ replaces $P(D_t)$ by $P(D_t - i\Omega)$. Factor

$$P(z - i\Omega) = z^m Q(z), \qquad Q(0) \neq 0.$$

The operator D_t^m is surjective on polynomials, increasing the required degree by at most m. On each finite-dimensional polynomial-degree filtration, $Q(D_t) = Q(0)(1 + N)$ with N nilpotent, hence has the finite inverse $Q(0)^{-1} \sum_r (-N)^r$. For a matrix operator, unimodular Smith transformations preserve $\mathscr{X}_{\text{ep}}$ and reduce the assertion to its scalar invariant factors. □

The complete output ledger after Noether restriction and local Diff $\times$Weyl $\times\, U(1)$ reduction is as follows. Here p and q are respectively the extra and Einstein characteristic polynomials; "zero factors" means persistent factors in the $\mathscr{X}_{\text{ep}}$ cokernel, not roots of a nonzero characteristic polynomial.

Output stratum	Reduced operator or invariant factors	Zero factors	Cokernel interpretation
Generic $L \geq 2$, nonstatic	$1, 1, p, pq$; off-shell inverse and Lemma 5.1 on p/q shells	0	none
Generic $L \geq 2$, static	$p = -(3\Lambda + 3\kappa^2 - 2)/3 < 0$ and $q > 0$ for $\Lambda \geq 6$	0	none
Exceptional $L = 1$, nonstatic	invertible off shell; nonzero factors on $\Omega^2 - K^2 = 4, 4/3$	0	none in $\mathscr{X}_{\mathrm{ep}}$
Scalar $L = 0$, $(\Omega, K) \neq (0, 0)$	exact modulo local gauge and Noether identities	0	none
Homogeneous $L = 0$, $(\Omega, K) = (0, 0)$	constraints plus $D_t^4(C - K)$ and $D_t^2 A_x$	2	ζ_H, ζ_{P_x}
Axial dipole $L = 1$, $(\Omega, K) = (0, 0)$	twist polynomial operator; polar block invertible	3	$\zeta_{J_1}, \zeta_{J_2}, \zeta_{J_3}$
Remaining certified global/twist outputs	contained in the preceding $L = 0, 1$ blocks	0 additional	none additional

For completeness, put $\kappa := K_{\mathrm{out}}$ for the output circle momentum (unrelated to the homogeneous function $K(t)$ below). The static generic entries use

$$p = -\frac{3\Lambda + 3\kappa^2 - 2}{3} < 0, \qquad q = \Lambda^2 + 2\Lambda\kappa^2 - 2\Lambda + \kappa^4.$$

Putting $s = \Lambda - 6 \geq 0$ and $u = \kappa^2 \geq 0$ gives

$$q = s^2 + 2su + u^2 + 10s + 12u + 24 > 0,$$

so no static $L \geq 2$ zero factor is hidden in the table.

For the homogeneous source, quadratic products have degree at most six. The two dynamical invariants therefore have the explicit polynomial right inverses

$$D_t^4\left(\frac{t^{r+4}}{(r+1)(r+2)(r+3)(r+4)}\right) = t^r, \qquad D_t^2\left(\frac{t^{r+2}}{(r+1)(r+2)}\right) = t^r, \quad 0 \leq r \leq 6.$$

This displays rather than merely asserts the zero-block surjectivity.

Theorem 5.2 (Complete finite-support exponential-polynomial cone)**.** *For the complete carrier V_{fin},*

$$\boxed{\mathcal{Z}_2^{\mathrm{ep}} = \{u \in V_{\mathrm{fin}} : \mu_H(u) = \mu_{P_x}(u) = \mu_{J_1}(u) = \mu_{J_2}(u) = \mu_{J_3}(u) = 0\}.} \tag{10}$$

Every point of this cone admits a real, spatially periodic second-order correction in $\mathscr{X}_{\mathrm{ep}}$, and each of the five equations is necessary.

Proof. Finite input support and Clebsch–Gordan selection produce finitely many output labels $j = (L, M, K, \Omega, \text{parity})$, with temporal polynomial degree at most six. The table exhausts those labels. Lemma 5.1 removes every nonzero invariant factor, including every nonzero-frequency resonance. The displayed polynomial primitives remove the homogeneous dynamical rows. Exactly five zero factors remain, so

$$\operatorname{coker} \mathcal{L}_{\mathrm{ep}} = \operatorname{span}\{\zeta_H, \zeta_{P_x}, \zeta_{J_1}, \zeta_{J_2}, \zeta_{J_3}\}.$$

Pairing the second-order equation with them and applying Proposition 3.1 proves necessity. Conversely, when all five moment maps vanish, every output source lies in the blockwise image. The finite corrections and their conjugates assemble a real element of $\mathscr{X}_{\mathrm{ep}}$, proving sufficiency. ☐

Remark 5.3. The theorem covers arbitrary finite sums across the certified generic, exceptional, homogeneous, and twist blocks. It is not a theorem for the full smooth, Sobolev, distributional, infinite-mode, or causal function space.

6 Complete bounded-obstruction decomposition

On $\mathscr{X}_{\mathrm{qp}}$, differentiation cannot produce a positive power of t, and a resonant pure exponential has no bounded quasiperiodic primitive. For each finite output block j, let $\mathcal{P}_{j,r}$ extract the coefficient of $t^r e^{-i\Omega_j t}$ for $r > 0$. After all such coefficients vanish, let $\mathcal{R}_{j,a}$ pair the remaining pure exponential with an exact basis of the reduced left kernel of $\mathcal{L}_j(-i\Omega_j)$. This definition includes the dynamical reduced left kernel at $\Omega_j = 0$. The stabilizer covectors inside the genuine zero block are split off and recorded separately as μ_X; no other zero-root covector is silently identified with them.

Theorem 6.1 (Complete finite bounded-obstruction ledger)**.** *For every $u \in V_{\mathrm{fin}}$, a second-order correction in $\mathscr{X}_{\mathrm{qp}}$ exists if and only if*

$$\mu_H(u) = \mu_{P_x}(u) = \mu_{J_i}(u) = 0, \qquad \mathcal{P}_{j,r}(u) = 0, \qquad \mathcal{R}_{j,a}(u) = 0 \tag{11}$$

for all finite output blocks and all corresponding indices. These are the complete linear obstruction types in the declared finite module. The common quadratic zero set of the resulting functionals is only partly classified.

Proof. If a bounded quasiperiodic v exists, then $\mathcal{L}v$ is also bounded and quasiperiodic, so every positive polynomial coefficient $\mathcal{P}_{j,r}$ vanishes. For each remaining pure-frequency block, finite-dimensional linear algebra says that $\mathcal{L}_j(-i\Omega_j)v_j = S_j$ is solvable precisely when S_j annihilates its reduced left kernel. At the genuine zero block those pairings split into the five Taub moment maps and any remaining dynamical zero-root functionals $\mathcal{R}_{j,a}$; at nonzero shells they are the finite-frequency $\mathcal{R}_{j,a}$. Off-shell blocks are invertible. Conversely, these conditions put every finite block source in the corresponding image, and the finite collection of solutions and conjugates assembles an element of $\mathscr{X}_{\mathrm{qp}}$. □

The certificates contain a moment-map-zero fixture with a nonzero $\mathcal{P}_{j,2}$ and another with a nonzero $\mathcal{R}_{j,a}$. Thus both new families are independent of the five global charges, and $\mathcal{Z}_2^{\mathrm{qp}} \subsetneq \mathcal{Z}_2^{\mathrm{ep}}$.

For finite generic oscillatory inputs the complete bounded $K = \Omega = 0$ receiver can be made fully explicit. Its nonzero source map is

$$(\mu_H, \mu_{P_x}, \mu_{J_1}, \mu_{J_2}, \mu_{J_3}, R_c), \qquad R_c = \frac{1}{2}\sum_j k_j^2 h_j.$$

Here h_j is the complete normalized Hermitian current block, including cross terms within a degenerate shell. The second formal homogeneous mean covector is the Wilson-acceleration row; its quadratic source vanishes by the integrated Maxwell identity. The polar $L = 1$ zero block contributes no physical cokernel, the axial block contributes exactly the three lifted rotations, and every static $L \geq 2$ block is reduced-invertible. Thus the six displayed conditions are necessary and sufficient for bounded inversion of the zero-frequency source. On the candidate-13 two-fibre carrier, joining them to the eighteen finite-frequency coefficients gives the complete formula

$$\mathcal{Z}_{2,13}^{\mathrm{qp}} = \{\mu_H = \mu_{P_x} = \mu_{J_1} = \mu_{J_2} = \mu_{J_3} = R_c = \mathcal{R}_{13,1} = \cdots = \mathcal{R}_{13,18} = 0\}.$$

Theorem 6.2 (Candidate-13 bounded-origin theorem)**.** *On the complete real candidate-13 generic two-fibre carrier,*

$$\mathcal{Z}^{\mathrm{qp}}_{2,13} = \{0\}.$$

The finite exponential-polynomial cone does not collapse: it remains the common zero of the five stabilizer moment maps and contains the explicit three-occupation Einstein–extra fixture.

Proof. Put

$$D = -\frac{8}{5L}\mu_H - \frac{3}{2L\sqrt{\rho}}\mu_{P_x} - \frac{2}{\rho}R_c, \qquad \rho = \frac{-250 + 461\sqrt{10}}{2132}.$$

In occupation variables this is

$$D = \frac{2}{5}\sum \omega^2 h - \frac{3}{8}\sum n\omega h - \sum n^2 h, \qquad n \in \{1, -2\}.$$

The Einstein-minus current is negative, while the extra and Einstein-plus current blocks are positive in both parities. Since $1/2 < \rho < 3/5$, exact rational bounds give positive coefficients at least 379/1000 and 23/80 on the two signed Einstein-minus fibres, and at least 101/250 and 38/45 on the corresponding extra fibres. The Einstein-plus frequencies are larger and their common positive-branch coefficient is increasing. Thus D is a strictly positive sum of the absolute-current Gram occupations for every parity and m. Vanishing of μ_H, μ_{P_x}, R_c forces every occupation, hence the tangent, to vanish. The rotations and eighteen resonance equations are then automatic. In the finite exponential-polynomial correction class R_c and the finite-frequency resonances have secular inverses, so only the five moment maps remain. □

The earlier three-occupation fixture has $R_c < 0$, consistently placing it in the finite exponential-polynomial cone but outside the bounded cone. The theorem is a real positivity result on this exact circumference; it makes no identification with another collision fibre and does not classify the complex zero variety.

7 Complete standard generalized-zero subcone

The coordinates in the next theorem are concrete perturbations, not abstract labels. In coordinates (t, x, θ, ϕ), write

$$h_{xx} = C(t), \qquad h_{\theta\theta} = K(t), \qquad h_{\phi\phi} = K(t)\sin^2\theta, \qquad a_x = W_x + Q_e t, \tag{12}$$

$$K(t) = a + bt, \qquad C(t) = at^2 + \frac{b}{3}t^3 + c + dt. \tag{13}$$

For an orthonormal real $\ell = 1$ basis (Y_{1a}, X_a), the twist block is

$$h_{xA} = (A_a + B_a t)X^a_A, \qquad a_x = -(A_a + B_a t)Y_{1a}. \tag{14}$$

Coordinate	Linear representative	Time behaviour	Geometric meaning
a	sphere scale $K = a$ with induced $C = at^2$	constant/quadratic	homogeneous radion/Jordan seed
b	$K = bt$, $C = bt^3/3$	linear/cubic	homogeneous Jordan partner
c	$C = c$	constant	circle circumference modulus
d	$C = dt$	linear	circumference velocity
Q_e	$a_x = Q_e t$	linear	uniform electric tangent
W_x	$a_x = W_x$, $\mathrm{d}a = 0$	constant	flat circle Wilson line
$A \in \mathbb{R}^3$	twist h_{xA}, a_x	constant	lifted $SO(3)$ mapping-torus holonomy
$B \in \mathbb{R}^3$	same twist carrier	linear	twist velocity/Jordan partner

Theorem 7.1 (Standard generalized-zero bounded cone)**.** *On the complete standard generalized-zero carrier with homogeneous coordinates (a, b, c, d, Q_e, W_x) and axial twist position/velocity (A, B), bounded second-order extension holds exactly on*

$$a = b = Q_e = 0, \qquad B = 0, \qquad c, d, W_x \text{ arbitrary}, \qquad A \in \mathbb{R}^3 \text{ arbitrary}. \tag{15}$$

The surviving source has an explicit bounded correction.

The triangular positive-degree ideal supplies the proof. The complete homogeneous positive-degree coefficients are

Equation row	coefficient of t	coefficient of t^2
E_{11}	$15ab$	$15b^2/2$
sphere trace	$-15ab/2$	$-15b^2/4$
Maxwell x row	$Q_e a$	$Q_e b/2$

The remaining leading polynomial tensor is $\mathrm{STF}(B \otimes B)t^2$, whose squared real coefficient is $2|B|^4/3$. They force $b = B = 0$ and $Q_e a = 0$. Intersecting with the five moment maps leaves

$$\mu_H = -(a^2 + Q_e^2), \qquad \mu_{P_x} = 0, \qquad \mu_{\mathbf{J}} = 0,$$

and hence forces $a = Q_e = 0$ on this carrier. The remaining homogeneous source vanishes, while constant twist position is tangent to an exact lifted mapping-torus family and has a constant polar correction.

Explicitly, on the universal cover identify

$$(x, p) \sim (x + L, \exp(A^a J_a)p), \qquad p \in S^2, \tag{16}$$

and lift the rotation to the magnetic bundle with its patchwise Maxwell compensator. The local product fields descend by covariance. Differentiating this exact quotient family at $A = 0$ gives the constant twist representative; the stored quadratic correction is its constant polar $L = 2$ Taylor term.

Corollary 7.2 (Universal polynomial elimination)**.** *For every point of the complete finite-support bounded cone,*

$$b = 0, \qquad B = 0, \qquad Q_e a = 0.$$

Oscillatory products cannot cancel these zero-frequency polynomial rows.

8 Natural transport columns

Several apparently dangerous global–oscillator products are derivatives of exact families and can therefore be classified without solving each harmonic source independently.

Proposition 8.1 (Electric and Wilson columns)**.** *For every certified nonzero-frequency standard or extra primary, the electric-charge cross source has a bounded correction obtained by differentiating electromagnetic duality,*

$$F_\theta = \cos\theta F + \sin\theta \star_g F.$$

At the background,

$$h_{\mathrm{cross}} = 0, \qquad f_{\mathrm{cross}} = \star_{\bar{g}} f + (D_g \star)[h] F_{\mathrm{bg}}.$$

This is the mixed derivative at the purely magnetic background, not a claim that the finite duality orbit remains in one magnetic bundle component. Its sphere period vanishes for $\ell \geq 1$, so this mixed correction itself lifts globally as a fixed-bundle connection difference. Every Wilson–oscillator source vanishes because $\delta A = W_x dx$ has $\delta F = 0$ and the local action depends on the connection through F.

This removes the complete Q_e and W_x oscillator columns from the bounded frontier. It does not remove the separate homogeneous condition $Q_e a = 0$.

Theorem 8.2 (Complete circumference column)**.** *Let c vary the circle radius and let a certified oscillator have compact momentum k. The c-times-oscillator source admits a bounded correction if and only if*

$$c = 0 \qquad \text{or} \qquad k = 0.$$

For $k \neq 0$, the radius family gives a correction in $\mathscr{X}_{\rm ep}$ but none in $\mathscr{X}_{\rm qp}$.

Indeed, writing $R^2 = 1 + \eta c$,

$$\omega_R^2 = \frac{k^2}{R^2} + m_{\rm branch}^2, \qquad \partial_\eta \omega_R|_0 = -\frac{ck^2}{2\omega}.$$

Differentiating the exact family produces

$$\partial_\eta e^{-i\omega_R t}\big|_0 = \frac{ick^2}{2\omega} t e^{-i\omega t}.$$

To make the shell inference explicit, write the reduced matrix equation as $L(\omega_R, R)u_R = 0$, choose a left on-shell vector $u^\dagger$, and differentiate at $R = 1$. The Feynman–Hellmann identity is

$$\left\langle u^\dagger, (\partial_\eta L)u \right\rangle = -\omega_0' \left\langle u^\dagger, (\partial_\omega L)u \right\rangle = \frac{ck^2}{2\omega} \left\langle u^\dagger, (\partial_\omega L)u \right\rangle. \tag{17}$$

The last factor is the current-normalized shell form. It is nonzero on each standard branch and nondegenerate on every extra multiplicity block. Hence the particular circumference source, not merely some source, has nonzero adjoint pairing whenever $ck \neq 0$ and the corresponding mode coefficient is nonzero. This is a genuine $\mathcal{R}_{j,a}$ obstruction, not a polynomial $\mathcal{P}_{j,r}$ obstruction. At $k = 0$, $\omega_0' = 0$ and differentiating the covariant circle indices gives an ordinary same-frequency correction.

9 Exceptional resonance as an independent obstruction

Moment-map cancellation does not guarantee bounded or even nonsecular second-order closure in a restricted exceptional carrier. For the axisymmetric exceptional $\ell = 1, k = 0$ axial and polar amplitudes (a_x, a_p), a twist velocity can cancel the time-translation moment map. The two parity self-sources can also be tuned to cancel their even $L = 2$ resonance. But the required interior point has $a_x a_p \neq 0$, and the mixed axial $L = 2$ source has exact nonzero adjoint pairing

$$-\frac{8\sqrt{3}}{9}.$$

The boundary axes retain their individual bounded resonant self-obstructions. Thus the only axisymmetric two-polarization exceptional configuration in this carrier admitting a correction in $\mathscr{X}_{\rm qp}$ is $a_x = a_p = 0$. Nonzero configurations on the moment-map cone still admit the secular $\mathscr{X}_{\rm ep}$ corrections guaranteed by Theorem 5.2. The example proves that bounded resonant cokernel equations contain information independent of global charges; it does not contradict the exponential-polynomial theorem.

10 A complete tuned nonzero-momentum all-primary cone

The preceding exceptional result is a no-go on a restricted carrier. The next fixture shows that a bounded mixed cone can instead survive after every primary at the same angular and compact-momentum fibre is admitted. Fix

$$\ell = 2, \qquad m_A = 0, \qquad k^2 = 2\sqrt{3} - \frac{7}{6}, \tag{18}$$

choosing the circle circumference so that $\pm k$ are allowed compact momenta. The three positive input frequencies are

$$\omega_-^2 = \frac{29}{6}, \qquad \omega_e^2 = 2\sqrt{3} + \frac{25}{6}, \qquad \omega_+^2 = \frac{29}{6} + 4\sqrt{3}, \qquad \omega_- < \omega_e < \omega_+. \tag{19}$$

Here q_- and q_+ denote the two Einstein primaries, while X denotes the complete extra-primary multiplicity. Include both momentum signs, both axial and polar q_- amplitudes, arbitrary normalized X and q_+ multiplicities, and one constant twist position. No nonaxisymmetric input is included.

Write $A_\pm, P_\pm$ for the axial and polar q_- coefficients at momenta $\pm k$. The exact 140-row collision census covers every output $L = 1, 2, 3, 4$, momenta $K = 0, \pm 2k$, and every zero, sum, and difference frequency of q_-, X, and q_+. Its unique characteristic hit is the polar extra shell

$$L = 4, \qquad K = 0, \qquad \Omega = 2\omega_-, \tag{20}$$

generated only by the q_-q_- product. Its two independent adjoint pairings are nonzero multiples of

$$A_+A_- - 3P_+P_-, \qquad A_+P_- - A_-P_+. \tag{21}$$

Their complete complex zero set consists of the two mixed sheets

$$A_\pm = \sigma\sqrt{3}\, P_\pm, \qquad \sigma \in \{+1, -1\}, \tag{22}$$

and the two one-sided planes $A_+ = P_+ = 0$ or $A_- = P_- = 0$.

On a mixed sheet define the nonnegative Einstein-minus occupations

$$N_\pm = -h_-(A_\pm e_A + P_\pm e_P, A_\pm e_A + P_\pm e_P), \tag{23}$$

and let $E_\pm$ and $B_\pm$ be the total nonnegative occupations in the extra and Einstein-plus multiplicities. The complete Hamiltonian and circle momentum equations are

$$\omega_e^2(E_+ + E_-) + \omega_+^2(B_+ + B_-) = \omega_-^2(N_+ + N_-), \tag{24}$$

$$\omega_e(E_+ - E_-) + \omega_+(B_+ - B_-) = \omega_-(N_+ - N_-). \tag{25}$$

Theorem 10.1 (Tuned axisymmetric all-primary bounded cone)**.** *In the carrier just declared, a bounded finite-quasiperiodic second-order correction exists precisely at the origin or on one of the two mixed sheets* (22) *with*

$$\frac{1 - r_e}{1 + r_e} \le \frac{|P_+|^2}{|P_-|^2} \le \frac{1 + r_e}{1 - r_e}, \qquad r_e := \frac{\omega_-}{\omega_e}, \tag{26}$$

and with nonnegative positive-primary occupations satisfying (24)–(25)*. Their phases and multiplicity factorizations are otherwise arbitrary. The declared constant twist coordinate remains free and bounded-removable.*

Proof. Equations (21) give the four components displayed above. On either one-sided plane, positivity and $r_e < 1$ force the negative occupation to vanish when $\mu_H = \mu_{P_x} = 0$; hence only the origin survives. On a mixed sheet, the raw axial and polar weights cancel. At fixed positive-branch energy, the largest possible compact momentum is supplied by the lower positive frequency ω_e, so (24)–(25) have a nonnegative solution if and only if

$$|N_+ - N_-| \le r_e(N_+ + N_-), \tag{27}$$

which is equivalent to (26). Sufficiency is explicit: set $B_+ = B_- = 0$ and choose

$$E_+ + E_- = r_e^2(N_+ + N_-), \qquad E_+ - E_- = r_e(N_+ - N_-). \tag{28}$$

The interval makes both $E_\pm$ nonnegative. The five stabilizer pairings and the unique nonzero-shell adjoint pairings then vanish. Every other entry in the exact collision ledger is off shell, while the nonzero-Fourier scalar complex and constant-twist cross column are boundedly removable. Theorem 6.1 therefore supplies a bounded correction. Necessity follows from the same complete obstruction ledger. □

Because $\omega_- < \omega_e < \omega_+$, the interval (26) strictly contains the interval obtained when only the Einstein-plus primary is allowed to balance the negative branch. Thus the extra primary creates no new resonance at this fixture but does enlarge the bounded nonlinear cone. This is a theorem for one tuned $\ell = 2, m_A = 0, |k|$ fibre, not for nonaxisymmetric modes, other harmonics or circumferences, or multiple absolute-momentum fibres.

11 All-angular-momentum promotion and the two-momentum workload

11.1 The tuned standard-branch cone for every $\ell \ge 2$

The standard Einstein-minus calculation admits a uniform angular-momentum promotion. For each integer $\ell \ge 2$, tune the compact momentum by

$$k^2 = \sqrt{2\ell(\ell+1)} - \frac{\ell}{2} - \frac{1}{6}. \tag{29}$$

Let $a_\pm$ and $p_\pm$ denote the axial and polar Einstein-minus amplitudes at momenta $\pm k$. The complete standard-branch collision census has one relevant opposite-momentum characteristic shell. Up to nonzero exact normalizations $R_\ell > 0$ and $X_\ell \ne 0$, its two adjoint pairings are

$$\mathcal{R}_{\text{polar}} = R_\ell \left(a_+ a_- - \frac{\ell(\ell+1)}{2} p_+ p_- \right), \tag{30}$$

$$\mathcal{R}_{\text{axial}} = X_\ell \left(a_+ p_- - a_- p_+ \right). \tag{31}$$

Their complex zero variety consists of two mixed sheets

$$a_\pm = \sigma \sqrt{\frac{\ell(\ell+1)}{2}}\, p_\pm, \qquad \sigma \in \{+1, -1\}, \tag{32}$$

and the two one-sided planes on which one momentum sign vanishes.

Theorem 11.1 (All-ℓ tuned axisymmetric standard cone)**.** *For every $\ell \geq 2$ at the tuning (29), the complete bounded quasiperiodic cone on the declared axisymmetric standard Einstein-minus/Einstein-plus carrier is the origin together with the two mixed sheets (32) satisfying*

$$\frac{1-r_\ell}{1+r_\ell} \leq \frac{|p_+|^2}{|p_-|^2} \leq \frac{1+r_\ell}{1-r_\ell}, \qquad r_\ell := \frac{\omega_-}{\omega_+}. \tag{33}$$

The one-sided resonance planes meet the Hamiltonian and circle-momentum balance equations only at the origin.

The proof is uniform in ℓ: the exact collision census excludes every other standard-branch shell, equations (30)–(31) give the full resonance ideal, and the two global charge equations reduce to the elementary occupation interval (33). This is a genuine all-ℓ theorem, but it is still separately tuned at each ℓ, axisymmetric, and confined to the standard branches. It does not join different angular momenta or different absolute compact momenta.

11.2 Two absolute momenta: a branch-basis obstruction census

The first two-absolute-momentum enlargement has closed all 164 declared branch-basis coefficients. The axisymmetric $L = 4$ calculation is complete: all 108 coefficients across axial–axial, polar–polar, and the two ordered cross-parity sectors are exact. The nonaxisymmetric $L = 3$ calculation contributes 44 exact coefficients and the $L = 1$ companion the remaining 12. Every declared individual branch-basis fixture in these two blocks has nonzero cokernel and a bounded obstruction.

This census by itself is only an exhaustive coefficient statement for the named basis columns: different columns may cancel after amplitudes are joined. The candidate-13 fibre is the first case where the arbitrary-amplitude ideal, the zero-frequency receiver, and the correction-class-dependent range have all been joined, giving the complete formula displayed after Theorem 6.1. The other collision circumferences admit a sharp scalar classification.

Theorem 11.2 (Collision scalar-separation classification)**.** *On every positive-ρ generic carrier with signed momenta $(1, -2)$, the common zero of μ_H, μ_{P_x}, R_c is the origin. Consequently the complete bounded generic cone is $\{0\}$ at collision candidates (1) through (15). At each same-sign candidate (16) through (21), the scalar common zero is nontrivial and no strict separator in $\operatorname{span}(\mu_H, \mu_{P_x}, R_c)$ exists. The six full resonance-joined bounded cones are not classified by this scalar theorem.*

Proof. For the opposite-sign carrier let (t_1, t_2) be the frequency midpoints between the Einstein-minus and extra branches at absolute momenta one and two. The shell polynomial associated with $(a, b, c) = (1, t_2/2 - t_1, -t_1 t_2/2)$ factors as

$$Q_1(\omega) = (\omega - t_1)(\omega + t_2/2), \qquad Q_{-2}(\omega) = (\omega - t_2)(\omega + 2t_1).$$

The second factors are positive. The first factors are negative on the negative-current Einstein-minus branch and positive on the extra and Einstein-plus branches. After the current sign is included, the resulting charge/pressure functional is a strictly positive sum of all absolute-current occupations. For each of the six same-sign collision values, the certificate instead gives four positive exact cofactors y_i satisfying

$$\sum_i y_i s_i(\omega_i^2, n_i\omega_i, n_i^2) = 0.$$

Farkas duality excludes a strict scalar separator, while the same positive weights give a nonzero scalar common-zero occupation. Finite-frequency resonances are separate conditions and are not inferred to vanish. □

Theorem 11.3 (Sharp collision nonemptiness split). *On each of the six distinct same-sign collision backgrounds (16)–(21), the complete generic bounded second-order cone contains a nonzero real point. Together with the preceding theorem, the twenty-one collision fibres split sharply as*

$$\mathcal{Z}^{(i)}_{2,\mathrm{bounded}} = \{0\} \quad (1 \le i \le 15), \qquad \mathcal{Z}^{(i)}_{2,\mathrm{bounded}} \neq \{0\} \quad (16 \le i \le 21).$$

This is a nonemptiness theorem for the last six cones, not a classification of their full real algebraic geometry.

Proof. For each same-sign fibre, use the four positive exact Farkas cofactors as absolute-current occupations and take axisymmetric input, so all three lifted rotation moment maps vanish. The same cofactors set $\mu_H = \mu_{P_x} = R_c = 0$. All 108 nonzero-frequency same-fibre temporal channels are off shell: the exact census contains 864 nonzero target-shell defects, while its $L = 0$ rows use the separately empty nonzero-Fourier and homogeneous nonzero-frequency quotients.

It remains only to kill the one cross-fibre resonance assigned to each candidate. On (17)–(19), the Farkas support omits one required resonant factor. On (16) and (20), choose the occupied modes axial with $m = 0$; the relevant $L = 3$ or $L = 1$ coefficient contains $\langle 2, 0; 2, 0 \mid L, 0\rangle = 0$. On (21), put the resonant $q_+(1), q_-(2)$ pair on the certified real mixed-parity $L = 4$ component $A_{\mathrm{pol}} = rA_{\mathrm{ax}}$, $B_{\mathrm{pol}} = sB_{\mathrm{ax}}$. Its two real nonzero parity vectors have positive absolute-current norms and may be scaled independently to the prescribed occupations. The complete finite-harmonic cokernel criterion then supplies a bounded finite-quasiperiodic second-order correction on each of the six backgrounds. □

Corollary 11.4 (Universal scalar cone and bounded section). *For every $\rho > 0$ on a same-sign $n = (1, 2)$ carrier, the nonnegative (μ_H, μ_{P_x}, R_c)-null occupation cone has exactly four extreme rays. Each contains both Einstein-minus nodes and chooses one of p_{extra}, q_+ independently on each momentum fibre. On each of the six collision backgrounds (16)–(21), all four scalar extreme rays admit a nonzero bounded amplitude lift. More strongly, every point of each complete four-ray scalar cone has a bounded amplitude lift. Thus the bounded cone projects surjectively onto the scalar occupation cone on all six backgrounds.*

Proof. Divide the scalar receiver column on branch b and fibre n by the positive factor n^2, and put $x_{b,n} = \omega_{b,n}/n = \sqrt{\rho + \mu_b/n^2}$. The columns lie on $(x^2, x, 1)$, with ρ-independent order

$$q_-^{(2)} < p^{(2)} < q_+^{(2)} < q_-^{(1)} < p^{(1)} < q_+^{(1)}.$$

The only non-obvious gap is $(9 - 5\sqrt{3})/2 > 0$, which follows from $81 > 75$. The current signs in this order are $(-, +, +, -, +, +)$. Four ordered moment-curve columns have the unique dependence

$$z_i = \frac{1}{\prod_{j \neq i}(x_i - x_j)}, \qquad \sum_i z_i x_i^d = 0 \quad (d = 0, 1, 2),$$

whose signs alternate. Hence exactly four supports agree with the current signs, proving the extreme-ray classification.

For the 24 lifts, ten supports omit a resonant factor and ten use the axisymmetric odd-L zero. The remaining four use real $L = 4$ components: two on candidate (19)'s regular-pencil eigenline and two on candidate (21)'s mixed-parity component. Independent fibre rescaling matches the extreme-ray occupations. This proves ray saturation only; cross terms between different ray lifts can reactivate the bilinear resonance map.

For occupation-surjectivity, do not add fixed ray representatives. Instead, given an arbitrary point of the scalar cone, choose all occupied vectors axial and axisymmetric on candidates (16), (17), (18), and (20). The odd- L coefficient then vanishes independently of the six occupation values. On (19) and (21), place the two resonant vectors on one fixed real mixed $L = 4$ component and rescale the two fibres independently to their prescribed absolute-current norms; choose all spectator vectors axial with $m = 0$. This defines a section on every cone face, including pairwise ray sums and the origin. It does not imply that every amplitude in the fibre over a given occupation is bounded. $\square$

Proposition 11.5 (Exact same-sign bounded fibre products). *For each candidate $i = 16, \ldots, 21$, let π_i be the six-node absolute-current occupation map, C_i its four-ray (μ_H, μ_{P_x}, R_c)-zero cone, μ_J the three lifted rotation moment maps, and B_i the complete isolated cross-fibre resonance map. Then the bounded or finite-quasiperiodic second-order tangent cone is exactly*

$$\mathcal{Z}_i^{\text{bnd}} = \pi_i^{-1}(C_i) \cap \mu_J^{-1}(0) \cap \mathcal{V}(B_i).$$

All relative phases, both parities and all $m = -2, \ldots, 2$ coefficients are retained.

Proof. Necessity is the complete finite-harmonic adjoint-cokernel theorem: the scalar factor gives H, P_x, R_c, the rotation factor gives the three lifted rotations, and $\mathcal{V}(B_i)$ kills the unique candidate-specific cross-fibre shell. Conversely, the 864-defect same-fibre census makes every other nonzero-frequency source block invertible, so simultaneous vanishing is sufficient. This gives an exact equation, not yet a decomposition of its real Hermitian components or singular strata. $\square$

Proposition 11.6 (Resonance fibres over the scalar-cone faces). *On each same-sign collision background (16)–(21), the complete complex positive-frequency phase/parity resonance fibre is stratified over the four-ray scalar cone. The two q_- occupations are strictly positive at every nonzero scalar-cone point. The automatic and active strata are*

candidate	*resonant pair*	*automatic face*	$\dim_{\mathbb{C}} \mathcal{V}_{\text{active}}$	$\# \operatorname{Irr}_{\mathbb{C}} \mathcal{V}_{\text{active}}$
16	$q_-^{(1)} q_-^{(2)}$	$\{0\}$	12	1
17	$q_-^{(1)} q_+^{(2)}$	$\text{cone}(R_1, R_2)$	14	1
18	$p^{(1)} q_-^{(2)}$	$\text{cone}(R_2, R_4)$	22	1
19	$p^{(1)} q_-^{(2)}$	$\text{cone}(R_2, R_4)$	10	4
20	$q_+^{(1)} q_-^{(2)}$	$\text{cone}(R_1, R_3)$	14	1
21	$q_+^{(1)} q_-^{(2)}$	$\text{cone}(R_1, R_3)$	10	2

Candidates (17) and (20) use their isolated difference-frequency channel; the other four rows use the isolated sum-frequency channel. On an automatic face one bilinear factor vanishes. Off it, the table lists the complete active all-m complex resonance variety. The four components on (19) are the real-supported regular-pencil eigenline components, and the two on (21) are the real mixed parity-proportionality components. Every scalar-cone point has at least one real representative in the corresponding stratum.

Proof. Every extreme ray contains $q_-^{(1)}$ and $q_-^{(2)}$ with positive weight. Reading the optional resonant node from the four ray supports gives the automatic faces in the table. On their complements, restrict the complete binary-form zero-variety decompositions: the target-doublet first-transvectant variety for (16), the third-transvectant kernels for (17),(20), the multiplicity-two first-transvectant variety for (18), the regular pencil for (19), and the scalar highest-spin product for (21). Their certified dimensions and component counts give the last two columns. The scalar-cone section in the preceding corollary supplies real points. Combined with the exact fibre-product formula, these strata are intersected with the prescribed real norm levels and the three lifted rotation moment maps to give a necessary-and-sufficient bounded equation. The proposition does not decompose that real intersection into connected components or singular strata. $\square$

Proposition 11.7 (Connected automatic-face rotation links)**.** *For candidates (17)–(21), fix a nonzero occupation in any relative support stratum of the automatic two-ray face in the preceding table. The complete amplitude link satisfying*

$$B_i = 0, \qquad \mu_{J_1} = \mu_{J_2} = \mu_{J_3} = 0$$

is nonempty and connected. Candidate (16) has no nonzero automatic face, so this statement is not applicable there. This is a fixed-occupation theorem; it neither glues distinct occupation strata nor classifies the active resonance varieties.

Proof. On an automatic face, one complete factor of the bilinear map B_i vanishes. For each occupied branch–momentum node, fix its positive absolute-current norm. The amplitude space is then a product of odd spheres. Quotienting by the independent node phases gives a compact connected product of complex projective spaces. The action-derived Lee–Wald form descends to a signed product of Fubini–Study forms. A negative q_- factor reverses orientation but remains nondegenerate.

The diagonal lifted $SO(3)$ action is Hamiltonian, and its moment map is the descended triple $(\mu_{J_1}, \mu_{J_2}, \mu_{J_3})$. Compactness makes this moment map proper. The connected-fibre theorem of Lerman, Meinrenken, Tolman and Woodward [9] therefore makes its zero fibre connected. The certified axisymmetric representative proves nonemptiness. Finally, its inverse image in the product of norm spheres has the connected node-phase torus as fibre, and is therefore connected. □

Proposition 11.8 (The axisymmetric section is rotation-critical)**.** *On the certified all-$m = 0$ section over every scalar-cone point of each candidate (16)–(21), the Jacobian of the three lifted rotation moment maps has rank zero at the origin and exactly two at every nonzero point. Hence zero is not a regular value along this section, and these explicit bounded points cannot seed a codimension-three implicit-function chart.*

Proof. In the spin-two basis $m = -2, -1, 0, 1, 2$, let e_0 denote the $m = 0$ vector. The exact invariant angular form gives

$$T_3 e_0 = 0, \qquad W_2 T_1 e_0 \text{ and } W_2 T_2 e_0 \quad \text{real-linearly independent.}$$

Thus $d\mu_{J_3} = 0$ on every occupied axisymmetric block, giving rank at most two. At a nonzero section point, at least one block has nonzero norm; variations supported in that block retain the two independent T_1, T_2 covectors with a nonzero current coefficient, giving rank at least two. At the origin all three differentials vanish because the moment map is quadratic. In particular, the connected automatic-face links above may contain critical points: connectedness does not imply regularity. The next local gate is the quadratic normal form of the missing J_3 equation on the kernel of the two transverse differentials. □

Proposition 11.9 (Automatic-face hyperbolic normal form)**.** *Fix a nonzero axisymmetric section point on any automatic-face support stratum of candidates (17)–(21), and let N be the number of occupied current eigenlines. On the angular slice aligned with those eigenlines, the quadratic μ_{J_3} form restricted to*

$$\ker d(\mu_{J_1}, \mu_{J_2})$$

has real inertia

$$(n_+, n_-, n_0) = (4N - 2, 4N - 2, 2).$$

In particular it is indefinite. Moreover, through every such point there is an exact fixed-occupation nonaxisymmetric curve inside the complete bounded fibre product. Candidate (16) has no nonzero automatic face, so the statement is not applicable there.

Proof. The spin-two angular form and J_3 form in the ordered basis $m = -2, -1, 0, 1, 2$ are

$$W_2 = \operatorname{diag}\left(1, \frac{1}{4}, \frac{1}{6}, \frac{1}{4}, 1\right), \qquad W_2 T_3 = \operatorname{diag}\left(-2, -\frac{1}{4}, 0, \frac{1}{4}, 2\right).$$

Rescale each node by the square root of its nonzero current magnitude and swap the $m = +1$ and $m = -1$ variables on negative-current nodes. This normalizes all coefficients simultaneously. Put $u_j = \delta z_{j,+1}$ and $v_j = \overline{\delta z_{j,-1}}$. The two real transverse rotation equations combine into one complex equation

$$\sum_{j=1}^{N} a_j(u_j + v_j) = 0, \qquad a_j > 0.$$

With $w_j = u_j + v_j$ and $z_j = u_j - v_j$, the $m = \pm 1$ part of μ_{J_3} is proportional to $\operatorname{Re} \sum_j w_j \overline{z_j}$. The constraint places w in $a^\perp$. Its pairing with the $a^\perp$ part of z gives real inertia $(2N-2, 2N-2)$, while the complex component of z parallel to a is a two-real-dimensional radical. The unrestricted $m = \pm 2$ coordinates contribute $(2N, 2N, 0)$, proving the displayed inertia.

For the exact curve, choose any occupied eigenline, fix its phase so that its $m = 0$ coefficient is $a > 0$, and replace its angular vector by

$$z(t) = \sqrt{a^2 - 12t^2}\, e_0 + te_{+2} + te_{-2}, \qquad 12t^2 < a^2.$$

Its norm is fixed because $(a^2 - 12t^2)/6 + t^2 + t^2 = a^2/6$. No adjacent magnetic weights are simultaneously occupied, so $\mu_{J_1} = \mu_{J_2} = 0$, while the two endpoint terms cancel in μ_{J_3}. The absent resonant node stays zero on an automatic face, hence $B_i = 0$ remains exact. This proves the curve lies in the necessary-and-sufficient bounded fibre product. The proposition does not classify internal current-orthogonal directions or the two-dimensional radical, and therefore is not a complete singular-stratum theorem. □

Corollary 11.10 (Complete fixed-norm rotation Hessian)**.** *At the same axisymmetric points, let N be the number of occupied scalar branch–momentum nodes and let D be the sum of their complex internal current-space dimensions. A q_- or q_+ node has $D_j = 2$, one axial and one polar current eigenline, whereas a p-extra node has $D_j = 4$, two eigenlines in each parity. On the complete tangent to the fixed-node-norm link, restricted by $\mathrm{d}(\mu_{J_1}, \mu_{J_2}) = 0$, the real inertia of the quadratic μ_{J_3} form is*

$$(4D - 2, 4D - 2, 2D - N + 2).$$

After quotienting the N independent node phases it is

$$(4D - 2, 4D - 2, 2D - 2N + 2).$$

Thus every ray and two-ray relative-interior stratum of the five nonzero automatic faces remains a transverse rotation saddle after all axial and polar internal directions are restored. Candidate (16) *remains not applicable.*

Proof. The aligned block is the preceding proposition. A current-orthogonal internal eigenline does not enter the two linear transverse-rotation equations. Its five complex angular coefficients see the exact weighted J_3 diagonal

$$\left(-2, -\frac{1}{4}, 0, \frac{1}{4}, 2\right),$$

and hence add real inertia $(4, 4, 2)$. Summing over the $D - N$ orthogonal eigenlines gives angular-kernel inertia $(4D - 2, 4D - 2, 2)$. The $m = 0$ fixed-node-norm tangent is radical of real dimension $2D - N$, which yields the first formula. The node-phase torus lies in that radical and removes N real directions, yielding the second. The dimensions are respectively $10D - N - 2$ and $10D - 2N - 2$, equal to the sums of the displayed inertia triples. Exact enumeration of all fifteen ray/interior support strata gives phase-reduced inertias ranging from $(30, 30, 10)$ to $(54, 54, 20)$.

This is a complete quadratic Hessian theorem at fixed node norms, not a classification of the nonlinear zero set along its radical. It neither glues occupation strata nor treats active resonance components. □

Proposition 11.11 (Candidate-(16) active restricted current)**.** *Fix both nonzero $q_-^{(1)}$ and $q_-^{(2)}$ current norms on candidate (16), and quotient their independent phases. The irreducible active resonance variety is a complex projective tenfold in $\mathbb{CP}^9 \times \mathbb{CP}^9$. On every complex smooth stratum, the restricted Lee–Wald form is nondegenerate and negative Kähler. On the generic smooth locus its real symplectic rank is* 20*.*

Proof. Before projectivization, the exact target-doublet $L = 3$ resonance equations factor into two rank-at-most-one determinantal cones. Their product is irreducible of complex dimension 12 in the 20-dimensional affine input space. The equations are bihomogeneous in the two nonzero nodes, so quotienting their two complex scalings gives the stated projective dimension.

Both resonant nodes are on the q_- branch. The complete axial/polar current is therefore negative definite on each node. Writing its descended Hermitian form as

$$h = -c_1 H_1 \oplus -c_2 H_2, \qquad c_1, c_2 > 0,$$

gives $h(v, v) < 0$ for every nonzero complex tangent vector on every smooth stratum. Its imaginary part is consequently nondegenerate there. The projective variety itself is singular, so this proposition does not apply a smooth-orbifold connected-fibre theorem to its full rotation-zero set. □

Proposition 11.12 (Active linear-sheet rotation links)**.** *Each of the four real pencil-eigenline components on candidate (19) and the two real parity-proportional components on candidate (21) has a nondegenerate restricted Lee–Wald current. Its resonant core has Hermitian inertia* $(5, 5, 0)$*. At every fixed nonzero active occupation, each component has a nonempty connected lifted-rotation zero link, including arbitrary occupied spectator nodes. The four and two components remain distinct.*

Proof. Every declared sheet is the direct sum of one complex five-dimensional subspace of a positive-current node and one complex five-dimensional subspace of a negative-current node. Each ambient node form is definite, so its restriction to a nonzero linear graph or pencil eigenline remains definite. Spectral orthogonality between distinct branch–momentum nodes then gives inertia $(5, 5, 0)$; occupied spectators add definite orthogonal blocks and cannot create a radical.

After fixing the two resonant-node norms and quotienting their phases, the core is $\mathbb{CP}^4 \times \mathbb{CP}^4$ with one positive and one negative weighted Fubini–Study factor. Spectators add projective factors. The diagonal lifted $SO(3)$ action preserves each internal graph or eigenline and is Hamiltonian. Its zero fibre is nonempty by choosing the spin-two $m = 0$ vector in every factor. Compactness makes the moment map proper, so the connected-fibre theorem gives a connected zero fibre; restoring the connected node-phase torus preserves connectedness. Distinct parity ratios and pencil eigenlines are unchanged by node phases, and the active nonzero-norm condition prevents passage through a one-sided component. $\square$

Proposition 11.13 (Candidate-(17)/(20) axisymmetric Zariski currents)**.** *At every nonzero active scalar-cone point on the certified all-axial $m = 0$ sections of candidates (17) and (20), the affine Zariski tangent to the complete third-transvectant resonance variety has complex dimension* 16 *and restricted Hermitian current inertia*

$$(6, 10, 0).$$

The section is algebraically singular because the irreducible variety has complex dimension 14*. Nevertheless, the restricted current is nondegenerate. After removing the two node scalings, the projective Zariski tangent has inertia* $(5, 9, 0)$ *and real symplectic rank* 28*.*

Proof. For the axisymmetric binary quartic $e_0 = (0, 0, 1, 0, 0)$, the exact third-transvectant matrix has rank two and a three-dimensional kernel. The two linearized parity equations therefore have total rank four in the twenty-dimensional affine input space, giving the stated tangent dimension.

The only possible current radical is controlled by equality of the absolute currents on the positive and negative resonant nodes. Exact outward rational intervals prove the strict opposite inequality on both active extreme rays of each scalar cone:

$$\mathsf{n}(q_-^{(1)}) > \mathsf{n}(q_+^{(2)}) \quad \text{on candidate (17)}, \qquad \mathsf{n}(q_-^{(2)}) > \mathsf{n}(q_+^{(1)}) \quad \text{on candidate (20)}.$$

Every inactive ray adds only to the negative occupation, so the inequalities hold throughout the active cones. In each parity channel, the three kernel directions give three positive/negative pairs and the two constrained directions are negative, yielding $(3, 5, 0)$. Doubling parity gives $(6, 10, 0)$. The two complex node-scaling directions have inertia $(1, 1, 0)$, and their removal gives $(5, 9, 0)$.

This is a Zariski-tangent theorem at singular points. It does not classify the restricted current on the full smooth locus or justify a connected-fibre argument there. $\square$

Proposition 11.14 (Smooth current radicals on candidates (17) and (20))**.** *Each complete active third-transvectant variety contains a smooth bounded point, with all five stabilizer moment maps zero, at which the fixed-norm projective Lee–Wald current has a nonzero radical. Consequently neither active variety is globally symplectic, and the smooth symplectic-orbifold connected-fibre theorem cannot be applied globally.*

Proof. The exact parity pencil has coefficient matrix $C_0 = \left(\begin{smallmatrix} a & b \\ b & 3a \end{smallmatrix}\right)$, with $b/(a\lambda) = \epsilon\sqrt{3}$, $\epsilon = \pm 1$, and $\lambda^2 = 128/5$. Put

$$D = \operatorname{diag}(1, 3), \qquad Q = \begin{pmatrix} \epsilon\sqrt{3} & -\epsilon\sqrt{3} \\ 1 & 1 \end{pmatrix}, \qquad P^T = (C_0 Q)^{-1}.$$

Then $Q^T D Q = 6I$, while

$$P = Q \operatorname{diag}\left(\frac{1}{6a(\lambda+1)}, -\frac{1}{6a(\lambda-1)}\right).$$

Thus the second transform is not a scalar multiple of the first. After the independent channel rescaling

$$S = \operatorname{diag}\left(\frac{3}{2}a(\lambda+1), \frac{3}{2}a(\lambda-1)\right), \qquad PS = Q \operatorname{diag}(1/4, -1/4),$$

both eigenchannels have normalized positive-to-negative current coefficient ratio $1/16$.

In either channel take

$$f = (1, 0, 0, 0, 1), \qquad g = (1, 0, 1, 0, 1).$$

The third-transvectant equations vanish and their Jacobian has rank three, so this is a smooth point with seven-dimensional affine tangent. Restricting $-H \oplus H/16$, where $H = \mathrm{diag}(1, 1/4, 1/6, 1/4, 1)$, gives rank six and the exact radical

$$\delta f = (0, 1/4, 0, 1/4, 0), \qquad \delta g = (0, 1, 0, 1, 0).$$

It is tangent to both norm levels and orthogonal to both phase directions, so it survives projectivization. Since $H(f, f) = 2$ and $H(g, g) = 13/6$, its positive-to-negative occupation ratio is $13/192$. Exact positive mixtures $R_3 + sR_1$ on candidate (17) and $R_2 + sR_1$ on candidate (20) realize that ratio inside their scalar cones. The reflection-symmetric representatives and their spectators have zero angular moments, while the scalar-cone equations kill the time and circle moments. The exact bounded fibre-product theorem then supplies the stated bounded points.

This is the explicit bounded point used below to calibrate the complete smooth determinantal locus. $\square$

Proposition 11.15 (Smooth current radicals on candidate (18))**.** *The complete candidate-(18) active resonance variety contains two exact families of smooth bounded points with all five stabilizer moment maps zero. On either family, the fixed-norm projective Lee–Wald current has a four-complex-dimensional radical. Hence this active variety is not globally symplectic either.*

Proof. Write X, Y for the current-orthogonal active functionals on the axial and polar p-extra doublets and U, V for the axial and polar q_- amplitudes. The exact $L = 3$ source pencil is equivalent to

$$T_1(X, U) + T_1(Y, V) = 0, \qquad 3T_1(X, V) + T_1(Y, U) = 0.$$

With

$$Q = \begin{pmatrix} \sqrt{3} & -\sqrt{3} \\ 1 & 1 \end{pmatrix}, \qquad P = Q^{-T},$$

one has $P^T Q = I$ and $P^T \left(\begin{smallmatrix} 0 & 3 \\ 1 & 0 \end{smallmatrix}\right) Q = \mathrm{diag}(\sqrt{3}, -\sqrt{3})$. The resonance equations therefore factor into two rank-one conditions $T_1(f_+, g_+) = T_1(f_-, g_-) = 0$.

Let $w_x, w_y > 0$ be the exact quotient-current weights of X, Y, and let $h_- > 0$ be the absolute axial q_- weight. Direct reduction of the action-derived current gives

$$G_p = \begin{pmatrix} w_x/12 + w_y/4 & -w_x/12 + w_y/4 \\ -w_x/12 + w_y/4 & w_x/12 + w_y/4 \end{pmatrix}, qquad |G_-| = 6h_- I.$$

Thus $z_+ = (1, 1)$ and $z_- = (1, -1)$ are exact eigenlines of G_p, with eigenvalues $w_y/2$ and $w_x/6$, respectively. On either line set $f = z_\pm \otimes e_0$, $g = t_\pm z_\pm \otimes e_0$, where $e_0 = (0, 0, 1, 0, 0)$ and

$$t_+^2 = \frac{w_y/2}{6h_-}, qquad t_-^2 = \frac{w_x/6}{6h_-}.$$

Both entries of $z_\pm$ are nonzero, so both rank-one factors are at smooth nonzero proportional pairs. For every $r \perp e_0$,

$$\delta f = z_\pm \otimes r, \qquad \delta g = t_\pm z_\pm \otimes r$$

is tangent and current-orthogonal to the complete tangent. These four complex directions are tangent to both norm levels and transverse to the two phase directions, so they survive projectivization.

Finally, exact rational intervals put the active occupation ratio below one on R_1 and above one on R_3. The unique positive mixture $R_3 + s_{18} R_1$ therefore gives equal $p^{(1)}_{\text{extra}}$ and $q^{(2)}_{-}$ absolute-current occupations, precisely the scaling above. The $m = 0$ resonant carriers and spectators kill the three rotation moments, while the scalar cone kills the time and circle moments. The exact bounded fibre-product theorem completes the claim.

This proves the two aligned branches used below to calibrate the complete smooth determinantal locus. □

Proposition 11.16 (Smooth presymplectic divisors and quotients)**.** *On the complete smooth active resonance varieties of candidates* (17), (18), (20)*, let J denote the full-row-rank constraint Jacobian and H the invertible ambient Lee–Wald Hermitian current. Then*

$$K = JH^{-1}J^{\dagger}$$

has

$$\ker K \cong \operatorname{rad}(H|_{\ker J}), \qquad \lambda \longmapsto H^{-1}J^{\dagger}\lambda.$$

Consequently $\det K = 0$ *is the complete smooth current-degeneracy locus, the determinantal ideals of K classify its higher-corank strata, and* $\ker J/\operatorname{rad}$ *carries an induced nondegenerate Hermitian current.*

For candidates (17) *and* (20)*, each parity factor has*

$$K_3(f,g) = J_3(f,g)H_3^{-1}J_3(f,g)^{\dagger}, \qquad H_3 = -H_{\text{ang}} \oplus H_{\text{ang}}/16,$$

and the two-parity divisor is $\det K_3(f_+, g_+) \det K_3(f_-, g_-) = 0$*. At the smooth bounded point of the preceding proposition,*

$$K_3 = \begin{pmatrix} 24 & 0 & 24 \\ 0 & 30 & 0 \\ 24 & 0 & 24 \end{pmatrix},$$

so its nullity is one. At the second exact smooth point

$$f = (-2, -2, -2, -2, -1), \qquad g = (12, 12, 11, 9, 0),$$

the same conormal determinant is 8293671904*. The degeneracy determinant is therefore not identically zero, so this is a proper smooth locus.*

For candidate (18)*, the same formula on every regular rank-one chart gives a chart-independent divisor. On the aligned section, writing $r = |t|^2$ and $b = 6h_- > 0$, its internal conormal factor satisfies*

$$\det C(r) = \frac{(2br - w_y)(6br - w_x)}{b^2 w_x w_y}.$$

The two factors are the symmetric and antisymmetric branches above. Since $3w_y - w_x > 0$, each is a simple internal branch with full conormal nullity four. Including the ten nondegenerate current-orthogonal spectators, the affine presymplectic quotient has complex dimension 18 *there.*

Proof. If $v \in \ker J$ is radical, nondegeneracy of H implies $Hv = J^{\dagger}\lambda$ for a unique conormal coefficient λ. The tangent equation is then $0 = Jv = JH^{-1}J^{\dagger}\lambda = K\lambda$. Conversely every $\lambda \in \ker K$ gives such a radical vector. Full row rank of J makes the displayed map injective. This proves the general claim and its quotient statement.

For the third-transvectant factors, direct exact substitution of the three constraint rows and H_3 gives the displayed matrix at the bounded point. For candidate (18), use a regular chart

$F_i = f_\alpha g_i - f_i g_\alpha$, $i \neq \alpha$, on each rank-one factor. Chart changes act on J by an invertible conormal matrix and hence on K by congruence, preserving its zero locus and nullity. At the aligned point the angular conormal factor is invertible, while the internal factor is $C(r) = rG_p^{-1} - I/b$ up to invertible diagonal congruence. Its determinant is the displayed factorization. The certified strict inequality $3w_y - w_x > 0$ makes the complementary eigenvalue nonzero on either branch, so tensoring with the four-dimensional angular conormal factor gives nullity four. The spectator statement follows from their certified current-orthogonal positive complement. □

Proposition 11.17 (Complete complex singular carrier for candidates (17) and (20))**.** *For one parity factor, write*

$$K_{T_3} = \{(f,g) \in \mathbb{C}^5 \oplus \mathbb{C}^5 : T_3(f,g) = 0\}.$$

Its Jacobian has rank below three precisely on

$$\mathrm{Sing}(K_{T_3}) = \left\{ \left(a\, v(\lambda), b\, v(\lambda)\right) : [\lambda_0 : \lambda_1 : \lambda_2] \in \mathbb{P}^2, (a,b) \in \mathbb{C}^2 \right\},$$

where

$$v(\lambda) = \left(3\lambda_2^2, -3\lambda_1\lambda_2, \lambda_0\lambda_2 + 2\lambda_1^2, -3\lambda_0\lambda_1, 3\lambda_0^2\right).$$

This singular locus has complex dimension four, is smooth away from the origin, and has projectivization $\mathbb{P}^2 \times \mathbb{P}^1$ embedded by $\mathcal{O}(2,1)$. Its incidence resolution is

$$\mathrm{Tot}(\mathcal{O}_{\mathbb{P}^2}(-2) \oplus \mathcal{O}_{\mathbb{P}^2}(-2)) \longrightarrow \mathrm{Sing}(K_{T_3}).$$

For the two parity factors, the complete singular locus has exactly two irreducible components,

$$(S_+ \times K_-) \cup (K_+ \times S_-),$$

each of complex dimension eleven, meeting in the dimension-eight product $S_+ \times S_-$.

Proof. A nonzero left-Jacobian covector $\lambda = (\lambda_0, \lambda_1, \lambda_2)$ gives

$$\lambda^T J(f,g) = \left(-(B_\lambda g)^T, (B_\lambda f)^T\right),$$

where B_λ is skew. Its determinant vanishes, while the three selected rank-four minors are

$$9\lambda_0^4, \qquad (\lambda_0\lambda_2 + 2\lambda_1^2)^2, \qquad 9\lambda_2^4.$$

If the outer coordinates vanish, the middle expression is $4\lambda_1^4$, so B_λ has rank four for every nonzero λ. Its kernel is generated by $v(\lambda)$. Thus a Jacobian rank loss is equivalent to f and g being proportional to this common kernel vector. The parametrization and incidence resolution follow. For the product, the Jacobian is block diagonal; hence it loses rank exactly when either factor does, giving the two components and their intersection.

This is a complex-algebraic theorem before real norm levels, node phases, lifted rotations, or Lee–Wald reduction. □

Corollary 11.18 (Candidate-(17)/(20) singular-component images intersect)**.** *At every positive active-occupation pair on candidate* (17) *or* (20)*, the two algebraic singular-component images meet in the node-phase and lifted rotation-zero quotient. Consequently their component labels supply no quotient-separation lower bound.*

Proof. The components

$$\Sigma_+ = S_+ \times K_-, \qquad \Sigma_- = K_+ \times S_-$$

meet in $S_+ \times S_-$. Put

$$f = \sqrt{6N_-}\, e_0, \qquad g = \sqrt{96N_+}\, e_0$$

in one parity factor and set the other factor to zero. The occupied pair is a common-square singular pair and the zero pair lies in the second singular cone, so this point is in the intersection. The exact current weights give occupations N_- and N_+. Its real completion is axisymmetric, hence rotation-zero, and both total node norms are positive, so both node phases act freely. The orbit of this point belongs to both quotient images.

This proves incidence, not connectedness within either image or of their union. □

Proposition 11.19 (Connected candidate-(17)/(20) double-singular hub)**.** *At every positive active-occupation pair on candidate* (17) *or* (20)*, the complete double-singular intersection $S_+ \times S_-$, after both common node-phase quotients, has a connected lifted-rotation moment-map zero fibre. Its compact incidence resolution is Kähler of complex dimension six.*

Proof. For either parity factor, the common-square incidence resolution is

$$\mathrm{Tot}(\mathcal{O}_{\mathbb{P}^2}(-2) \oplus \mathcal{O}_{\mathbb{P}^2}(-2)) \longrightarrow S.$$

Taking the product over $B = \mathbb{P}^2_+ \times \mathbb{P}^2_-$, fixing the positive and negative node norms, and dividing their two free phases gives

$$\mathbb{P}(L_+ \oplus L_-)_{\text{negative}} \underset{B}{\times} \mathbb{P}(L_+ \oplus L_-)_{\text{positive}}, \qquad L_\pm = \mathcal{O}_{\mathbb{P}^2_\pm}(-2).$$

This is a compact connected Kähler manifold of complex dimension $4 + 1 + 1 = 6$. The construction is equivariant for the lifted diagonal SO(3) action. Its Hamiltonian moment-map zero fibre is nonempty: the axisymmetric common-square representative used in Corollary 11.18 supplies a point at every positive occupation pair. Kirwan connectedness therefore makes the resolved zero fibre connected. The resolution is surjective onto the target double-singular hub and has connected fibres, so the target zero fibre is its connected continuous image.

The proposition concerns only the common hub. It does not assert that every component of $S_+ \times K_-$ or $K_+ \times S_-$ meets it. □

Proposition 11.20 (Frequency-weighted common-square quotient)**.** *Fix positive active occupations and restrict candidate* (17) *or* (20) *to one declared common-square parity factor, with the other factor at its vertex. After the two free node phases, the carrier is $\mathbb{CP}^2$. Put*

$$\delta = \omega_+ N_+ - \omega_- N_-.$$

If $\delta \neq 0$, the lifted-rotation zero set is $\mathbb{RP}^2$ and its SO(3) *quotient is one point. If $\delta = 0$, the complete $\mathbb{CP}^2$ is rotation-zero and its* SO(3) *quotient is a closed interval.*

Candidate (17) *has $\delta < 0$ throughout its complete nonzero active scalar cone. Candidate* (20) *has a nonempty exact balance divisor: $\delta < 0$ on R_2, $\delta > 0$ on R_4, and*

$$\left(-\frac{\delta_{R_4}}{\delta_{R_2}}\right) R_2 + R_4$$

has $\delta = 0$ with a strictly positive coefficient.

Proof. In descending binary-weight coordinates the common-square map is

$$[a, b, c] \longmapsto \left[a^2, ab, \frac{ac + 2b^2}{3}, bc, c^2\right].$$

The certificate checks equivariance against J_0, J_+, J_-. In the equivalent Cartesian Cartan model,

$$S(z) = zz^T - \frac{z^T z}{3} I \in \mathrm{Sym}^2_0(\mathbb{C}^3),$$

and direct multiplication gives

$$[S(z), S(z)^\dagger] = (z^\dagger z)\big(zz^\dagger - \overline{z}\, z^T\big).$$

Thus the Cartan-square moment map vanishes exactly when $[z]$ is phase-real. These directions form $\mathbb{RP}^2$, on which $\mathrm{SO}(3)$ is transitive.

The two occupied nodes have the same angular square direction, so their action-derived rotational moments combine as a positive normalization times δ and this Cartan-square moment map. At $\delta = 0$ the angular equation vanishes identically. Writing $z = x + iy$, a projective phase makes $x \cdot y = 0$; rotation and normalization then leave the complete orbit parameter

$$\eta = \frac{|z^T z|}{z^\dagger z} \in [0, 1].$$

Hence $\mathbb{CP}^2/\mathrm{SO}(3) \cong [0, 1]$.

Finally, exact substitution of the four circuit weights into $\omega_+ N_+ - \omega_- N_-$ gives negative signs on R_3, R_4 for candidate (17), negative sign on R_2 and positive sign on R_4 for candidate (20). Candidate-(17) inactive rays contribute only $-\omega_- N_-$. The displayed positive candidate-(20) combination therefore proves the balance claim.

This proposition concerns one parity endpoint of the connected hub. It does not classify either complete two-parity singular component. □

Proposition 11.21 (Balanced singular radial contraction)**.** *Fix positive active occupations on candidate* (17) *or* (20)*. On either complete singular component*

$$\Sigma_+ = S_+ \times K_-, \qquad \Sigma_- = K_+ \times S_-,$$

there is an occupation-preserving radial path from every point to the double-singular hub whose rotational residual is exactly

$$\mu_{\mathrm{rot}}(t) = (1 - t^2)\delta\, \mu_{\mathrm{sq}}, \qquad \delta = \omega_+ N_+ - \omega_- N_-.$$

Consequently, on candidate (20)*'s exact balance divisor $\delta = 0$, the complete fixed-occupation singular-union rotation-zero fibre is connected.*

For candidate (17)*, and for candidate* (20) *off balance, the same conclusion is certified only on the phase-real common-square sublocus $\mu_{\mathrm{sq}} = 0$.*

Proof. Consider first a point of $S \times K$. Write $A_\pm$ for the two node occupations carried by the common-square factor and $B_\pm$ for those carried by the arbitrary kernel factor. Scale the latter amplitudes by $t \in [0, 1]$, so their occupations become $t^2 B_\pm$, and add $(1 - t^2)B_\pm$ to the square factor along one common-square direction. The total occupations $N_\pm = A_\pm + B_\pm$ are unchanged. Scaling preserves the third-transvectant kernel equation, and the receiving factor remains common-square, so the complete bilinear resonance continues to vanish.

Let $\alpha = \omega_+ A_+ - \omega_- A_-$ and let M_K denote the initial rotational moment of the kernel factor. Initial rotation zero says

$$M_K = -\alpha\, \mu_{\mathrm{sq}}.$$

Along the path, the kernel contribution scales by t^2, whereas the square coefficient becomes $\alpha + (1-t^2)(\omega_+ B_+ - \omega_- B_-)$. Hence

$$\begin{aligned}\mu_{\mathrm{rot}}(t) &= t^2 M_K + \big[\alpha + (1-t^2)(\omega_+ B_+ - \omega_- B_-)\big]\mu_{\mathrm{sq}} \\ &= (1-t^2)(\omega_+ N_+ - \omega_- N_-)\mu_{\mathrm{sq}}.\end{aligned}$$

If the receiving square factor starts at its vertex, choose any square direction for $t < 1$. Its transferred amplitude vanishes as $t \to 1$; initial rotation zero gives $M_K = 0$, and the same formula supplies a continuous balance-divisor path. At $t = 0$ the arbitrary kernel factor is at its vertex, so the endpoint lies in $S_+ \times S_-$. The construction is symmetric under exchange of the two parity factors.

On candidate (20)'s balance divisor the displayed residual vanishes for every initial point. Every point of both singular components therefore joins the connected hub of Proposition 11.19, proving the claim. Off balance, $\mu_{\mathrm{sq}} = 0$ is sufficient for the same radial path. A nonzero residual obstructs this particular path only; it is not a nonradial no-go. □

Proposition 11.22 (Moving-square contraction and its sharp ansatz obstruction)**.** *On either candidate-*(17)/(20) *singular component, retain uniform scaling of the arbitrary K factor and exact transfer of its released occupations, but allow the receiving common-square direction to move continuously. Put*

$$\alpha = \omega_+ A_+ - \omega_- A_-, \qquad \delta = \omega_+ N_+ - \omega_- N_-.$$

Then every point with $\alpha\delta > 0$ contracts to the connected double-singular hub. The same holds for phase-real square directions and for the square-factor vertex.

For a non-phase-real point with $\delta \neq 0$, the remaining complementary declared ansatz is obstructed when $\alpha\delta < 0$: its receiving coefficient has one interior zero at which the scaled kernel moment is nonzero. If $\alpha = 0$, by contrast, every point contracts by a coefficient-zero square pre-rotation followed by the phase-real uniform contraction.

Proof. In the normalized Cartesian Cartan model, choose a projective phase so that $z = x + iy$, $x \cdot y = 0$, and $\|x\|^2 + \|y\|^2 = 1$. With $u = 2\|x\|\|y\|$, direct use of

$$[S(z), S(z)^\dagger] = \big(z^\dagger z\big)\big(zz^\dagger - \overline{z}\, z^T\big)$$

and $\|S(z)\|^2 = (2+u^2)/3$ gives the normalized moment radius

$$F(u) = \frac{3u}{2+u^2}, \qquad F'(u) = \frac{3(2-u^2)}{(2+u^2)^2} > 0 \quad (0 \le u \le 1).$$

Rotations supply every direction, so the complete normalized Cartan-square moment image is a closed ball. In particular, every multiple $r\mu_0$, $0 \le r \le 1$, is reached continuously from an initial moment μ_0.

Let $s = t^2$ be the remaining occupation fraction in the uniformly scaled kernel factor. The receiving square coefficient is

$$c(s) = s\alpha + (1-s)\delta.$$

Initial rotation zero gives $M_K = -\alpha\mu_0$. Hence the moving-square choice

$$\mu_{\mathrm{sq}}(s) = r(s)\mu_0, \qquad r(s) = \frac{s\alpha}{c(s)}$$

satisfies

$$-s\alpha\mu_0 + c(s)\mu_{\mathrm{sq}}(s) = 0.$$

If $\alpha\delta > 0$, then

$$\frac{dr}{ds} = \frac{\alpha\delta}{c(s)^2} > 0, \qquad r(0) = 0, \quad r(1) = 1,$$

so the Cartan moment ball supplies the entire path. At $s = 0$ the kernel factor is at its vertex and the receiving square is phase-real, placing the endpoint in the connected hub. The phase-real and square-vertex cases use zero square moment directly.

If $\alpha\delta < 0$, then

$$s_0 = -\frac{\delta}{\alpha - \delta} \in (0,1), \qquad c(s_0) = 0.$$

For nonzero μ_0, the remaining kernel moment $-s_0\alpha\mu_0$ is nonzero, so no square direction repairs this ansatz. If $\alpha = 0$, initial rotation zero instead gives $M_K = 0$. At $s = 1$ the square coefficient is also zero, so the Cartan moment-ball path may first move the square direction continuously from μ_0 to a phase-real direction without changing any occupation or rotational moment. Keeping that square moment zero while s then decreases from one to zero gives a second rotation-zero segment ending in the hub. This admissible pre-rotation was omitted by the former endpoint-continuity argument.

This classifies the stated ansatz, not independent node scaling or a general deformation of the K factor inside $T_3(f,g) = 0$. □

Proposition 11.23 (Independent-node bottleneck incidence). *Fix the two directions of the arbitrary K factor on either candidate-(17)/(20) singular component, but scale their squared amplitudes independently by $(x,y) \in [0,1]^2$, transferring the released occupations to the receiving common-square factor. Write*

$$a = \omega_- B_- > 0, \qquad b = \omega_+ B_+ > 0,$$

and let $U = a\mu_f$, $V = b\mu_g$ be the fixed weighted node moments. On a strict opposite-sign stratum $\alpha\delta < 0$, define

$$c(x,y) = \delta + ax - by, \qquad M_K(x,y) = -xU + yV,$$

and

$$\mathcal{I} = \{(x,y) \in [0,1]^2 : c(x,y) = 0,\ M_K(x,y) = 0\}.$$

Then the declared fixed-direction independent-node-scaling ansatz contracts to the connected double-singular hub if and only if $\mathcal{I} \neq \varnothing$.

If $U, V \neq 0$, incidence is possible only when $V = \kappa U$ for some $\kappa > 0$. In that case its unique candidate is

$$y_* = -\frac{\delta}{a\kappa - b}, \qquad x_* = \kappa y_*,$$

and it exists exactly when $a\kappa \neq b$ and $(x_*, y_*) \in [0,1]^2$. The one-zero cases are

$$U = 0,\ V \neq 0: \quad (x_*, y_*) = (-\delta/a, 0), \qquad U \neq 0,\ V = 0: \quad (x_*, y_*) = (0, \delta/b),$$

again subject to the displayed nonzero coordinate belonging to $[0,1]$. Thus non-positively-collinear nonzero node moments are obstructed within this ansatz, whereas every physical incidence point gives an explicit contraction.

Proof. Exact occupation transfer gives

$$A_-(x) = A_- + (1-x)B_-, \qquad A_+(y) = A_+ + (1-y)B_+,$$

so the receiving coefficient is $c(x,y)$, with

$$c(1,1) = \alpha, \qquad c(0,0) = \delta.$$

Independent nonnegative scaling preserves $T_3(f,g) = 0$, and the rotational equation is

$$M_K(x,y) + c(x,y)\mu_{\mathrm{sq}} = 0,$$

where μ_{sq} ranges over the closed Cartan moment ball proved in Proposition 11.22. Since $\alpha\delta < 0$, every continuous path from $(1,1)$ to $(0,0)$ meets $c = 0$. At such a point the rotational equation forces $M_K = 0$, proving necessity.

Conversely, choose $(x_*, y_*) \in \mathcal{I}$. Along the straight segment from $(1,1)$ to (x_*, y_*), affine linearity gives

$$c = (1-\tau)\alpha, \qquad M_K = (1-\tau)M_K(1,1).$$

The initial square direction therefore cancels this entire segment. Hold (x_*, y_*) fixed and move the square moment continuously to zero; this preserves the equation because $c = M_K = 0$. Finally, along the straight segment from (x_*, y_*) to $(0,0)$,

$$M_K = 0, \qquad c = \tau\delta,$$

so a fixed phase-real square direction completes the path to the hub.

For nonzero U, V, the equation $xU = yV$ with $x, y \geq 0$ has a nonzero solution precisely when U, V lie on the same positive ray. Substitution of $V = \kappa U$ into $M_K = c = 0$ gives the displayed formula and box criterion. If exactly one moment vanishes, the same two equations give the two boundary formulae directly. $\square$

Proposition 11.24 (Deformable-kernel component-incidence normal form)**.** *On a strict opposite-sign candidate-*(17)/(20) *stratum, let $F, G \in \mathbb{C}^5$ be the actual amplitudes of the two nodes in the arbitrary third-transvectant-kernel factor and put*

$$\overline{\mathcal{K}} = \{(F,G) : T_3(F,G) = 0,\ \|F\|_W \leq 1,\ \|G\|_W \leq 1\}.$$

Thus a vanished node carries no auxiliary projective direction. Let

$$\mathcal{M} = \overline{\mathcal{K}} / \big(U(1)_F \times U(1)_G \times SO(3)_{\mathrm{lift}}\big)$$

be the compact stratified semialgebraic orbit space, including its nonfree and algebraically singular strata. With

$$c(F,G) = \delta + a\|F\|_W^2 - b\|G\|_W^2, \qquad M_K(F,G) = -a\,m(F) + b\,m(G),$$

define

$$\mathcal{A} = \{[F,G] \in \mathcal{M} : \|M_K(F,G)\| \leq |c(F,G)|\}, \qquad \mathcal{I} = \{[F,G] \in \mathcal{A} : c(F,G) = 0 = M_K(F,G)\}.$$

Then a rotation-zero initial point contracts to the connected double-singular hub if and only if its path component in $\mathcal{A}$ meets $\mathcal{I}$.

The incidence is nonempty in both strict sign chambers. If $\delta < 0 < \alpha$, it contains the boundary point

$$G = 0, \qquad \|F\|_W^2 = -\delta/a, \qquad m(F) = 0.$$

If $\alpha < 0 < \delta$, it contains

$$F = 0, \qquad \|G\|_W^2 = \delta/b, \qquad m(G) = 0.$$

These statements apply separately to candidate (17) *and to the two candidate-*(20) *sign chambers; they do not identify their carriers or assert that every component of $\mathcal{A}$ meets $\mathcal{I}$.*

Proof. The receiving common-square moment has the complete closed unit ball as its image. Hence a square direction solving

$$M_K + c\mu_{\mathrm{sq}} = 0$$

exists exactly on $\mathcal{A}$. Compact real-algebraic invariant theory makes $\mathcal{M}$, $\mathcal{A}$, and their components semialgebraic; each connected component is semialgebraically path connected. The compact group slice theorem lifts paths in the orbit space across all stabilizer types.

For completeness, the required square-moment path lift is explicit. In the Cartan model write $z = x + iy$, use projective phase to impose $x \cdot y = 0$, and rotate the resulting oriented orthogonal frame. If $u \in [0, 1]$ is its norm ratio, the normalized moment radius and inverse branch are

$$r(u) = \frac{3u}{2 + u^2}, \qquad u(r) = \frac{3 - \sqrt{9 - 8r^2}}{2r}, \qquad u(0) = 0.$$

For $0 < r \leq 1$ the fibre is the connected rotation orbit about the moment axis; at $r = 0$ it is the connected phase-real $\mathbb{RP}^2$. Thus every semialgebraic path in $\mathcal{A}$ lifts from the declared initial square direction.

If a contraction exists, c changes from α to δ, so the intermediate-value theorem supplies a point with $c = 0$. Rotation zero then forces $M_K = 0$; hence the initial component meets $\mathcal{I}$.

Conversely, lift a path in that component to an incidence point. There $c = M_K = 0$, so the square direction is unconstrained and may be moved to a phase-real zero-moment direction. Now set

$$F_s = \sqrt{s}F_*, \qquad G_s = \sqrt{s}G_*.$$

Odd bilinearity preserves $T_3(F_s, G_s) = 0$, while

$$M_K(F_s, G_s) = sM_K(F_*, G_*) = 0, \qquad c(F_s, G_s) = (1 - s)\delta.$$

The phase-real square therefore completes the path to the hub.

Finally, $\alpha = \delta + a - b$. If $\delta < 0 < \alpha$, then $a > b - \delta > -\delta$, so $0 < -\delta/a < 1$. If $\alpha < 0 < \delta$, then $b > \delta + a > \delta$, so $0 < \delta/b < 1$. Phase-real spin-two directions have zero moment and give the two displayed boundary incidences. □

Proposition 11.25 (Candidate-(18) complete complex singular resolution)**.** *Let $R_\pm = \mathrm{Rank}_{\leq 1}(5 \times 2)_\pm$. The complete candidate-(18) complex carrier is*

$$\mathbb{C}^{10}_{\mathrm{spect}} \times R_+ \times R_-.$$

Its singular locus has exactly two irreducible components,

$$\Sigma_+ = \mathbb{C}^{10} \times \{0\} \times R_-, \qquad \Sigma_- = \mathbb{C}^{10} \times R_+ \times \{0\},$$

each of complex dimension sixteen. They meet in the dimension-ten spectator space. A smooth connected dimension-twenty-two resolution with connected fibres is

$$\mathbb{C}^{10} \times \mathrm{Tot}\big(\mathcal{O}_{\mathbb{P}^4}(-1)^{\oplus 2}\big) \times \mathrm{Tot}\big(\mathcal{O}_{\mathbb{P}^4}(-1)^{\oplus 2}\big)\,.$$

Proof. The ten 2×2 minors defining one factor have Jacobian rank four at a canonical nonzero rank-one matrix and rank zero at the vertex. The $GL(5, \mathbb{C}) \times GL(2, \mathbb{C})$ action is transitive on the nonzero rank-one locus, so the vertex is the complete singular locus of each factor. The standard incidence resolution

$$\mathrm{Tot}\big(\mathcal{O}_{\mathbb{P}^4}(-1)^{\oplus 2}\big) \longrightarrow R_\pm$$

is smooth and connected, with exceptional fibre $\mathbb{P}^4$. The product formula gives the two singular components, their intersection, and the global resolution. The ten positive current-orthogonal spectators remain explicit throughout.

This does not impose real fixed occupations, restrict the Lee–Wald current to the resolution, or reduce node phases and lifted rotations. □

Corollary 11.26 (Singular rotation-zero sections are unavoidable)**.** *For each candidate* (17)*,* (18)*, and* (20)*, and for every pair of positive active occupations $N_- > 0$, $N_+ > 0$, the real fixed-occupation carrier contains a singular point at which all three lifted rotational moment maps vanish. Both total node norms are nonzero, so the two node-phase actions remain free.*

Proof. Let $e_0 = (0, 0, 1, 0, 0)$. It is axisymmetric, so its real conjugate completion has vanishing rotational moment maps. For candidates (17) and (20), put

$$f = \sqrt{6N_-}\, e_0, \qquad g = \sqrt{96N_+}\, e_0$$

in one parity factor and set the other to zero. Since $e_0 = v([0 : 1 : 0])/2$, the occupied factor lies in the common-square singular locus and the other lies at its vertex. The exact current weights give the prescribed occupations.

For candidate (18), let

$$\alpha = \frac{w_x + 3w_y}{12} > 0, \qquad \beta = 6h_- > 0,$$

put

$$f = \sqrt{\frac{6N_+}{\alpha}}\, e_0, \qquad g = \sqrt{\frac{6N_-}{\beta}}\, e_0$$

in one rank-one factor, and set the other factor and all ten spectators to zero. This has the prescribed occupations and lies on a complete singular component. Positive total occupations make both common node-phase actions free in every case.

The corollary supplies an existence section, not a real component, connectedness, or singular-quotient classification. □

Proposition 11.27 (Candidate-(18) singular quotient has at least two components)**.** *At every fixed level with $N_- > 0$, the candidate-(18) singular locus is the disjoint union of two nonempty clopen subsets preserved separately by the two node phases and lifted $SO(3)$. Both subsets meet the rotational moment-map zero fibre. Consequently the singular rotation-zero quotient has at least two components.*

Proof. The two singular components are

$$\Sigma_+ = \mathbb{C}^{10} \times \{0\} \times R_-, \qquad \Sigma_- = \mathbb{C}^{10} \times R_+ \times \{0\},$$

and meet only in $\mathbb{C}^{10} \times \{0\} \times \{0\}$. The negative-current active occupation is carried by the g-columns of the two rank-one factors; the spectators are positive-current directions. On the intersection both g-columns vanish, so $N_- = 0$. Hence every $N_- > 0$ level separates the singular locus into two nonempty relatively closed-and-open pieces. The central sections of Corollary 11.26, with the occupied factor exchanged, give a rotation-zero point in each piece. Node phases and lifted rotations preserve the labelled vanishing-factor conditions, so no orbit meets both pieces.

This does not prove either quotient component connected or the full smooth-plus-singular rotation-zero fibre disconnected; smooth paths may still join the two singular pieces. □

Proposition 11.28 (Candidate-(18) smooth bridge between singular components)**.** *For every $N_- > 0$, $N_+ > 0$, one certified point in each candidate-(18) singular component is joined inside the fixed-occupation rotation-zero carrier by a path whose interior lies in the smooth complex carrier. Both node-phase actions remain free along the path.*

Proof. Let $w(\theta) = (\cos\theta, \sin\theta)$, $0 \le \theta \le \pi/2$, and let e_0 be the central angular vector. With

$$A = \begin{pmatrix} a & c \\ c & a \end{pmatrix}, \qquad a + c = \frac{w_y}{2} > 0, \quad a - c = \frac{w_x}{6} > 0, \qquad \beta = 6h_- > 0,$$

set the two positive and negative parity amplitudes to

$$u(\theta) = \sqrt{\frac{6N_+}{w(\theta)^T A w(\theta)}}\, w(\theta), \qquad v(\theta) = \sqrt{\frac{6N_-}{\beta}}\, w(\theta),$$

and multiply every column by e_0; set all spectators to zero. The two fixed occupations are constant by construction. Only $m = 0$ is occupied, so all rotational moment maps vanish. At the endpoints one rank-one factor vanishes, placing the path in the two different singular components. In the open interval both factors are nonzero rank one, so the complex carrier is smooth. Positive occupations keep both total node norms nonzero.

This joins the two distinguished singular sections, not every point or component of the complete rotation-zero fibre. □

Proposition 11.29 (Fixed-occupation node-phase-reduced divisors)**.** *On the regular nonzero-occupation locus, append to the smooth resonance Jacobian J the two Hermitian-orthogonality rows for the negative and positive physical nodes:*

$$\widehat{A} = \begin{pmatrix} J \\ C_- \\ C_+ \end{pmatrix}, \qquad \widehat{K} = \widehat{A} H^{-1} \widehat{A}^\dagger.$$

Then $\ker \widehat{A}$ *is a horizontal model for the two fixed norm levels modulo their common node phases, and*

$$\ker \widehat{K} \cong \operatorname{rad}\big(H|_{\ker \widehat{A}}\big)\,.$$

Thus $\det \widehat{K} = 0$ *is the complete regular current-degeneracy locus after the two node-phase quotients.*

For candidates (17) *and* (20)*,* $\widehat{A}$ *is* 8×20*. The two total-node rows couple the parity factors, so the reduced equation is one* 8×8 *determinant, not the product of the two affine* 3×3 *determinants. The bounded point above has* $\operatorname{rank} \widehat{K} = 6$*, horizontal complex dimension* 12*, reduced radical dimension* 2*, and linear quotient dimension* 10*. At*

$$f = (1, 1, 0, 0, 0), \qquad g = (1, 0, 0, 0, 0)$$

in both parity channels, $\det \widehat{K} = -3845153895680$*, so the reduced divisor is proper.*

For candidate (18)*, the ten positive current-orthogonal spectators are retained. The* 100 *product charts have* $\widehat{A} \in \mathrm{Mat}_{10\times 30}$ *and horizontal complex dimension* 20*. On the common central-angular section, set*

$$a = \frac{w_x}{12} + \frac{w_y}{4}, \qquad c = -\frac{w_x}{12} + \frac{w_y}{4}, \qquad b = 6h_-.$$

For real proportionality parameters t_1, t_2*,*

$$\det \widehat{K} = -\frac{128(t_1^2 + t_2^2)\left[a^2 - ab(t_1^2 + t_2^2) + b^2 t_1^2 t_2^2 - c^2\right]^4}{9b^7(a-c)^4(a+c)^3}.$$

The symmetric and antisymmetric aligned branches have $\operatorname{rank} \widehat{K} = 6$*, reduced radical dimension* 4*, and linear quotient dimension* 16*. The point* $(t_1, t_2) = (0, 1)$ *is nondegenerate.*

Proof. The fixed-norm tangent and phase gauge condition for each nonzero node combine into the single complex equation $C_\pm v = 0$. Hence $\ker \widehat{A}$ is the standard complex horizontal model. The conormal proof of Proposition 11.16, with J replaced by $\widehat{A}$, gives the radical isomorphism and determinantal criterion.

For candidates (17) and (20), exact substitution of the two third-transvectant Jacobians and the two total-node rows gives the stated ranks and determinant. For candidate (18), choose one nonzero entry of each rank-one 5×2 matrix. The four proportionality equations in each factor, the two node rows and all ten spectators give the stated 10×30 matrix. On chart overlaps its conormal rows change invertibly, so the augmented normal Grams are congruent. Direct determinant reduction gives the displayed factor. On the two aligned roots the augmented normal rank is six. Finally,

$$a^2 - ab - c^2 = \frac{w_x w_y - b w_x - 3b w_y}{12} < 0,$$

because the exact intervals give $w_x > 0$, $0 < w_y < 1$, while $\sqrt{3} > 17/10$ gives $b = 5760\sqrt{3} - 6912 > 2880$. This proves the nondegenerate control.

On every smooth constant-corank stratum, closedness of the reduced Lee–Wald form makes its radical distribution involutive. The resulting simple local leaf spaces are symplectic. This does not construct a global Hausdorff leaf space or perform the lifted $SO(3)$ reduction. □

Proposition 11.30 (Local lifted-rotation descent through current leaves)**.** *Let* U *be any smooth constant-corank fixed-occupation stratum of candidate* (17)*,* (18)*, or* (20)*, after the two common node phases, and let*

$$\mathcal{R} = \ker \Omega_U.$$

Every lifted rotational moment-map component is constant on the connected leaves of $\mathcal{R}$*. On a simple saturated neighbourhood* $q : U_0 \to \overline{U} = U_0/\mathcal{R}$*, there are unique* $\overline{\Omega}$ *and* $\overline{\mu}$ *satisfying*

$$q^*\overline{\Omega} = \Omega_U, \qquad q^*\overline{\mu} = \mu, \qquad d\langle \overline{\mu}, \xi \rangle = \iota_{\overline{\xi^\#}} \overline{\Omega}.$$

In particular,

$$\left(\mu^{-1}(0) \cap U_0\right)/\mathcal{R} \cong \overline{\mu}^{-1}(0).$$

Thus removing the Lee–Wald radical and imposing the lifted rotational constraint commute locally. The two operations remain distinct.

Proof. The lifted $SO(3)$ action preserves Ω_U, hence it preserves $\mathcal{R}$. Hamiltonianity gives, for every $r \in \mathcal{R}$,

$$d\langle \mu, \xi \rangle(r) = \Omega_U(\xi^\#, r) = 0.$$

Therefore μ is basic on connected radical leaves. Closedness and constant rank make Ω_U basic as well, so both descend uniquely to the simple local leaf space. Pulling back the descended Hamiltonian identity gives the original one, and surjectivity of q proves it. The zero-fibre identity follows immediately. Where the residual lifted $SO(3)$ action has a regular local quotient, this also identifies the two orders of local reduction.

This argument does not construct a global Hausdorff leaf space, prove connectedness of a complete rotation-zero fibre, resolve singular strata, or glue occupations. □

Proposition 11.31 (Candidate-(16) singular rotation-zero link)**.** *At every fixed pair of positive active-node occupations, the candidate-(16) projective active variety has exactly two disjoint singular endpoint strata, each isomorphic to $\mathbb{CP}^4$. Its complete lifted-$SO(3)$ moment-map zero fibre is nonempty and connected.*

Proof. Each transformed parity factor is the rank-at-most-one cone of 5×2 matrices. Its ten minors have Jacobian rank four at rank one and rank zero only at the vertex. On the positive two-node-norm link, the product singular locus therefore consists precisely of the two strata where one factor is the vertex; after the node-phase quotient each is $\mathbb{CP}^4$.

Resolve one factor by

$$\mathrm{Tot}(\mathcal{O}_{\mathbb{CP}^4}(-1) \oplus \mathcal{O}_{\mathbb{CP}^4}(-1)) \longrightarrow \mathrm{Rank}_{\le 1}(5 \times 2).$$

The product resolution has complex dimension 12. Fixing both positive node norms gives compact connected weighted $S^3 \times S^3$ fibres over $\mathbb{CP}^4 \times \mathbb{CP}^4$; quotienting the two free node phases gives a compact connected Kähler tenfold. The resolution is equivariant for the lifted diagonal $SO(3)$, and the imported axisymmetric bounded point makes its zero fibre nonempty. Kirwan connectedness makes that resolved zero fibre connected. Its continuous image is the complete singular target zero fibre, because the exceptional fibres are connected. No orbifold claim is used. □

Corollary 11.32 (Candidate-(16) occupation gluing)**.** *After normalizing total absolute-current occupation to one, the complete candidate-(16) active lifted-rotation zero link is connected over all nonzero scalar occupations.*

Proof. The three-dimensional scalar cone has four positive extreme rays. Its total-occupation-one section is a compact connected two-dimensional convex polytope, and both active q_- norms stay strictly positive. Projection of the complete rotation-zero link to this polytope is surjective by the exact bounded section and proper because its source is closed in a compact product. Every fibre is connected by Proposition 11.31. A proper surjection with connected fibres over a connected base has connected total space. □

12 Current bounded frontier

The electric, Wilson, and circumference columns are closed, and the first full-time velocity repair sharpens the a, d-frontier. Earlier direct d-source fixtures were evaluated at $t = 0$. They correctly proved that the constant shell-projection matrix is invertible, but they could not prove that the full source was bounded. No frozen paper theorem asserted the stronger conclusion; the intermediate "complete d-column" reading is withdrawn and replaced by the following full-time statement.

Locality gives

$$\mathcal{E}^{(2)}[dt, u](t) = t\,\mathcal{E}^{(2)}[c, u](t) + \mathcal{E}^{(2)}[dt, u](0).$$

For the two axial extra representatives and the first polar representative, the coefficient of t vanishes. For the second polar representative it is the nonzero eight-row vector

$$(-36, 0, 164, -8\sqrt{3}i, 0, -100, -24, -20).$$

Thus the isolated fixture has polynomial zero locus $dz_{\mathrm{polar},2} = 0$. The previously computed constant-shell adjoint matrix remains valid, but cannot by itself establish boundedness. Combining this restored term with the four direct radion-position columns gives the complete positive-degree cross ideal on the $\ell = 2, k = 0$ extra multiplet:

$$\langle az_{\mathrm{ax},1}, az_{\mathrm{ax},2}, az_{\mathrm{pol},1}, az_{\mathrm{pol},2}, dz_{\mathrm{pol},2} \rangle. \tag{34}$$

These five monomials are already a Gröbner basis in the declared amplitude order. Their zero locus has three faces:

1. all extra amplitudes zero, with a, d unrestricted by this cross ledger;
2. $a = 0$ and $z_{\mathrm{pol},2} = 0$, with d and the two axial plus first polar amplitudes free;
3. $a = d = 0$, with all four extra amplitudes free.

This is a complete $\mathcal{P}$-zero locus for the declared cross block, not by itself a complete bounded cone. Before the other conditions are imposed, its surviving faces must still satisfy moment maps, constant shell pairings, self-products, and twist cross terms. On the complete global-plus-extra carrier those remaining conditions now close as follows.

Theorem 12.1 (Complete global plus $\ell = 2, k = 0$ extra bounded cone)**.** *Adjoin all four axial/polar $\ell = 2, k = 0$ extra-primary amplitudes $(x_{\mathrm{ax},1}, x_{\mathrm{ax},2}, x_{\mathrm{pol},1}, x_{\mathrm{pol},2})$ to the complete homogeneous and twist carrier. The $\mathscr{X}_{\mathrm{qp}}$ tangent cone is exactly*

$$\mathcal{Z}^{\mathrm{qp}}_{2,\,\mathrm{global}+\ell 2X} = \{(c, d, W_x, A) : c, d, W_x \in \mathbb{R},\ \ A \in \mathbb{R}^3\}. \tag{35}$$

In particular, it contains no nonzero $\ell = 2, k = 0$ extra-primary direction.

Proof. The universal polynomial rows first force $b = B = 0$ and $Q_e a = 0$. The remaining Hamiltonian moment map is

$$\mu_H = -a^2 - Q_e^2 - \frac{4}{3}X, \tag{36}$$

$$X = 1296|x_{\mathrm{ax},1}|^2 + \frac{208}{3}|x_{\mathrm{ax},2}|^2 + 22464|x_{\mathrm{pol},1}|^2 + 12288|x_{\mathrm{pol},2}|^2. \tag{37}$$

It is a strictly negative sum of squares, so its real zero locus has $a = Q_e = x_{\mathrm{ax},1} = x_{\mathrm{ax},2} = x_{\mathrm{pol},1} = x_{\mathrm{pol},2} = 0$. What remains is precisely the standard cone of Theorem 7.1, whose bounded correction is already explicit. The repaired ideal (34) is automatically satisfied there. ☐

After these repairs, the still-open full-carrier work is:

1. standard Einstein oscillators and exceptional $\ell = 1$ inputs outside the separately classified fixtures, other harmonics, and generic nonzero momenta;
2. their a, d polynomial columns and shell-pairing zero loci;
3. shell resonances among bounded wave products and remaining mixed blocks; and

4. the common zero locus for arbitrary finite sums.

Any future completion must preserve the separation between positive-degree obstructions $\mathcal{P}$ and shell pairings $\mathcal{R}$.

Carrier or column	$\mathscr{X}_{\mathrm{ep}}$ correction	$\mathscr{X}_{\mathrm{qp}}$ correction
Complete finite-support cone	iff five moment maps vanish	additional $\mathcal{P}$ and $\mathcal{R}$ conditions
Standard generalized-zero block	classified	classified completely
Global plus $\ell = 2, k = 0$ extra	moment-map cone contains balanced extra directions	bounded cone equals (c, d, W_x, A); no extra wave
Electric charge $\times$ oscillator	finite quasiperiodic	removable by duality transport
Wilson $\times$ oscillator	zero source	zero source
Circumference $\times$ $k = 0$ mode	ordinary same-frequency	removable
Circumference $\times$ $k \neq 0$ mode	secular radius derivative	obstructed
Velocity $d\times$ selected polar extra mode	finite polynomial source	nonzero $\mathcal{P}_{j,1}$ unless canceled jointly
Exceptional axial–polar interior	secular primitive on moment cone	bounded-resonance obstructed
Tuned $\ell = 2, m_A = 0$, nonzero $\lvert k\rvert$, all primaries	moment-map cone	Theorem 10.1: two mixed sheets over the exact occupation polytope
$a, d \times \ell = 2, k = 0$ extra	allowed on moment cone	polynomial ideal (34), then $\mathcal{R}$ still required
Remaining a, d oscillator columns	allowed by ep theorem on moment cone	open

13 Interpretation

The solution set near a symmetric compact background is not generally a linear space. Its second-order approximation is a cone cut out first by a moment map and then, if uniform boundedness is demanded, by resonance conditions. Three points are important.

First, a linearly nonradical extra Weyl mode may fail to be tangent to the nonlinear fixed-charge solution space. This is nonlinear selection, not linear gauge removal. Second, mixing Einstein and extra directions can cancel the moment maps, so no conclusion follows from the sign of a pure branch alone. Third, even complete moment-map cancellation leaves bounded resonances. Exponential-polynomial formal continuation and bounded perturbative stability are therefore different claims.

The tuned nonzero-momentum theorem adds a complementary point: enlarging the positive branch can widen a surviving bounded cone without introducing a new shell collision. The extra primary is therefore neither uniformly deleted nor uniformly liberated by the quadratic test; its disposition depends on the complete carrier and correction class.

The action-normalized cubic fixture sharpens the same lesson. Its global compact-stabilizer class vanishes after the complete homogeneous-correction quotient, while bounded shell inversion fails for the certified second-order representative. Global charge cancellation and bounded resonant solvability are therefore independent tests at third order as well as at second order. Since the shell quotient by arbitrary second-order homogeneous additions has not been computed, this is not a no-go for every representative of the balanced two-jet.

The result supplies the middle map in the programme's residual atlas:

$$X_\alpha \longmapsto \Omega(X_\alpha, \overline{X}_\alpha) \longmapsto \mu(X_\alpha) \longmapsto D^2\mathcal{E}[X_\alpha, X_\alpha].$$

Interaction sourcing, detector response, and BRST status are separate later maps.

14 What is not proved

The following remain outside the theorem:

- infinite harmonic sums and Sobolev completion;
- retarded or advanced solvability and nonlinear causal dependence;
- integration of a second-order jet to an exact or all-orders family;
- a bounded third-order verdict after quotienting the shell functionals by every admissible homogeneous second-order addition;
- a third-order classification for arbitrary finite harmonic sums;
- the final residual/background-stabilizer quotient;
- asymptotically flat, de Sitter, anti-de Sitter, or black-hole boundary conditions;
- observer coupling, particles, quantum BRST cohomology, and unitarity.

These exclusions are not cosmetic. In particular, Theorem 5.2 does not establish nonlinear stability.

15 Certificate and reproduction ledger

The scoped mathematical claims in this manuscript are THEOREM-FROZEN for external specialist review. Its third-order source baseline is repository commit `aa7f7ff984e930f71e208538deac6e22b9cc22cf`.

Claim	Authoritative certificate
Complete finite-support exponential-polynomial cone and bounded ledger	`bridge/certificates/einstein_maxwell_weyl_complete_finite_harmonic_smooth_global_second_order.json`
Finite-generic bounded zero-frequency receiver	`bridge/certificates/einstein_maxwell_weyl_finite_generic_bounded_zero_block.json`
Complete candidate-13 bounded and smooth cone formulas	`bridge/certificates/einstein_maxwell_weyl_ell2_two_abs_momentum_candidate13_complete_mixed_cone.json`
Candidate-13 scalar-separation bounded-origin theorem	`bridge/certificates/einstein_maxwell_weyl_candidate13_scalar_separation_no_go.json`
All-collision scalar-separation classification	`bridge/certificates/einstein_maxwell_weyl_collision_scalar_separation_classification.json`
Same-sign collision same-fibre shell census	`bridge/certificates/einstein_maxwell_weyl_same_sign_collision_same_fibre_census.json`

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Claim	Authoritative certificate
Six same-sign collision bounded witnesses	`bridge/certificates/einstein_maxwell_weyl_same_sign_collision_bounded_witnesses.json`
Universal same-sign scalar extreme rays	`bridge/certificates/einstein_maxwell_weyl_same_sign_scalar_extreme_rays.json`
Candidatewise 120-minor/90-circuit scalar audit	`bridge/certificates/einstein_maxwell_weyl_same_sign_collision_scalar_occupation_cones.json`
All 24 same-sign scalar extreme-ray lifts	`bridge/certificates/einstein_maxwell_weyl_same_sign_extreme_ray_lifts.json`
Bounded sections over the complete same-sign scalar cones	`bridge/certificates/einstein_maxwell_weyl_same_sign_scalar_cone_sections.json`
Exact same-sign bounded phase/parity fibre products	`bridge/certificates/einstein_maxwell_weyl_same_sign_phase_parity_fibre_product.json`
Complete same-sign resonance fibres over scalar-cone faces	`bridge/certificates/einstein_maxwell_weyl_same_sign_resonance_face_fibres.json`
Connected fixed-occupation rotation-zero links on automatic faces	`bridge/certificates/einstein_maxwell_weyl_same_sign_automatic_face_rotation_links.json`
Axisymmetric rotation critical-locus theorem	`bridge/certificates/einstein_maxwell_weyl_same_sign_axisymmetric_rotation_singularity.json`
Automatic-face aligned rotation normal form and exact arcs	`bridge/certificates/einstein_maxwell_weyl_same_sign_automatic_face_rotation_normal_form.json`
Complete phase-reduced internal rotation normal form	`bridge/certificates/einstein_maxwell_weyl_same_sign_automatic_face_full_internal_rotation_normal_form.json`
Complete unquotiented and phase-quotiented fixed-norm rotation Hessians	`bridge/certificates/einstein_maxwell_weyl_same_sign_automatic_face_full_rotation_normal_form.json`
Candidate-(16) active restricted-current theorem	`bridge/certificates/einstein_maxwell_weyl_same_sign_candidate16_active_restricted_current.json`
Candidate-(16) singular strata and connected rotation-zero fibre	`bridge/certificates/einstein_maxwell_weyl_same_sign_candidate16_singular_rotation_zero_fibre.json`
Candidate-(16) nonzero occupation gluing	`bridge/certificates/einstein_maxwell_weyl_same_sign_candidate16_occupation_gluing.json`
Candidate-(19)/(21) active linear-sheet rotation links	`bridge/certificates/einstein_maxwell_weyl_same_sign_active_linear_sheet_rotation_links.json`
Candidate-(17)/(20) axisymmetric Zariski-tangent currents	`bridge/certificates/einstein_maxwell_weyl_same_sign_candidate17_20_axisymmetric_restricted_current.json`
Candidate-(17)/(20) smooth active-current radicals	`bridge/certificates/einstein_maxwell_weyl_same_sign_L1_active_restricted_current_degeneracy.json`
Candidate-(18) smooth active-current radical families	`bridge/certificates/einstein_maxwell_weyl_same_sign_candidate18_active_restricted_current_degeneracy.json`
Complete smooth active presymplectic divisors and tangent quotients	`bridge/certificates/einstein_maxwell_weyl_same_sign_active_presymplectic_divisors.json`
Complete candidate-(17)/(20) complex singular carrier	`bridge/certificates/einstein_maxwell_weyl_same_sign_third_transvectant_singular_locus.json`
Candidate-(18) complete complex singular resolution	`bridge/certificates/einstein_maxwell_weyl_same_sign_candidate18_complex_singular_resolution.json`
Unavoidable singular rotation-zero sections at every positive occupation	`bridge/certificates/einstein_maxwell_weyl_same_sign_active_singular_rotation_zero_sections.json`

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Claim	Authoritative certificate
Candidate-(18) singular quotient component lower bound	`bridge/certificates/einstein_maxwell_weyl_same_sign_candidate18_singular_component_separation.json`
Candidate-(18) smooth bridge between singular components	`bridge/certificates/einstein_maxwell_weyl_same_sign_candidate18_singular_smooth_bridge.json`
Candidate-(17)/(20) physical singular-component incidence	`bridge/certificates/einstein_maxwell_weyl_same_sign_candidate17_20_singular_component_incidence.json`
Candidate-(17)/(20) connected double-singular rotation-zero hub	`bridge/certificates/einstein_maxwell_weyl_same_sign_candidate17_20_double_singular_rotation_zero_fibre.json`
Candidate-(17)/(20) one-parity common-square rotation quotient	`bridge/certificates/einstein_maxwell_weyl_same_sign_candidate17_20_common_square_rotation_quotient.json`
Candidate-(17)/(20) singular radial contraction	`bridge/certificates/einstein_maxwell_weyl_same_sign_candidate17_20_singular_radial_contraction.json`
Candidate-(17)/(20) moving-square contraction and ansatz obstruction	`bridge/certificates/einstein_maxwell_weyl_same_sign_candidate17_20_moving_square_contraction.json`
Candidate-(17)/(20) independent-node bottleneck incidence	`bridge/certificates/einstein_maxwell_weyl_same_sign_candidate17_20_independent_node_scaling_contraction.json`
`bridge/certificates/einstein_maxwell_weyl_same_sign_candidate17_20_deformable_kernel_incidence_normal_form.json`	
Fixed-occupation node-phase-reduced divisors and local quotients	`bridge/certificates/einstein_maxwell_weyl_same_sign_active_phase_reduced_presymplectic_divisors.json`
Local lifted-rotation descent through current leaves	`bridge/certificates/einstein_maxwell_weyl_same_sign_active_local_rotation_leaf_descent.json`
Stabilizer action and lifted rotations	`bridge/certificates/einstein_maxwell_weyl_plebanski_hacyan_stabilizer.json`
Generalized-mode symplectic form	`bridge/certificates/einstein_maxwell_exceptional_global_symplectic.json`
Generalized-mode moment maps	`bridge/certificates/einstein_maxwell_weyl_exceptional_global_moment_maps.json`
Standard generalized-zero bounded cone	`bridge/certificates/einstein_maxwell_weyl_standard_global_bounded_second_order.json`
Electric and Wilson transport	`bridge/certificates/einstein_maxwell_weyl_electric_wilson_complete_oscillator_transport.json`
Complete circumference column	`bridge/certificates/einstein_maxwell_weyl_circumference_complete_oscillator_bounded_classification.json`
First full-time velocity polynomial column	`bridge/certificates/einstein_maxwell_weyl_d_ell2_extra_full_time_polynomial.json`
Repaired $a, d \times \ell = 2$ polynomial zero locus	`bridge/certificates/einstein_maxwell_weyl_ad_ell2_extra_polynomial_zero_locus.json`
Complete global plus $\ell = 2, k = 0$ extra bounded cone	`bridge/certificates/einstein_maxwell_weyl_complete_global_ell2_extra_bounded_cone.json`
Exceptional two-polarization resonance	`bridge/certificates/einstein_maxwell_weyl_exceptional_ell1_two_polarization_resonance.json`
Opposite-momentum exponential-polynomial theorem	`bridge/certificates/einstein_maxwell_weyl_opposite_momentum_smooth_global_second_order.json`

Continued on next page

Claim	Authoritative certificate
Tuned axisymmetric all-primary bounded cone	`bridge/certificates/einstein_maxwell_weyl_opposite_momentum_ell2_tuned_all_primary_bounded_cone.json`
All-ℓ standard-branch resonance and bounded cone	`bridge/certificates/einstein_maxwell_weyl_symbolic_ell_tuned_axisymmetric_bounded_cone.json`
Two-momentum axisymmetric $L = 4$ matrices	`bridge/certificates/einstein_maxwell_weyl_ell2_two_abs_momentum_axial_axial_L4_matrix.json` and the three parity companion matrices
Two-momentum nonaxisymmetric $L = 3$ matrix	`bridge/certificates/einstein_maxwell_weyl_ell2_two_abs_momentum_nonaxisymmetric_L3_matrix.json`
Two-momentum nonaxisymmetric $L = 1, 3$ completion	`bridge/certificates/einstein_maxwell_weyl_ell2_two_abs_momentum_nonaxisymmetric_L1_L3_matrix.json`
Moment-map/Taub bridge	`bridge/certificates/einstein_maxwell_weyl_moment_map_taub_bridge.json`
Balanced third-order global quotient and four-shell bounded obstruction	`bridge/certificates/EINSTEIN_WEYL_COMPACT_CAUCHY_THIRD_ORDER_KURANISHI_EVALUATION_V1.json`

Each certificate records its schema, imported hashes, exact commands, test tier, assumptions, and missing-object ledger. The independent verifier for each named certificate has passed before the scoped claims were marked THEOREM-FROZEN. This lifecycle state promotes only the complete formulas displayed above. The candidate-13 real bounded geometry and all fifteen opposite-sign collision carriers are closed. Each same-sign collision cone has a certified nonzero bounded point, while its full real component geometry remains open; so does the unrestricted finite-support bounded zero set.

16 Outlook

The immediate mathematical target is the full real component geometry of the six same-sign collision cones. Their nonemptiness, same-fibre shell census, universal four-ray scalar projection, all 24 extreme-ray lifts, and a bounded section over every scalar-cone point are now closed. The complete complex resonance fibre is also stratified over every scalar-cone face. The remaining same-sign gate has now split. On the five nonzero automatic faces, every fixed-occupation rotation-zero link is nonempty and connected. On all six candidates, however, the explicit axisymmetric section lies in the rotation critical locus with Jacobian rank two. The active resonance strata still require a restricted-current nondegeneracy test before any connected-fibre theorem can be applied, while the critical section requires the quadratic normal form of the missing rotation equation. On the automatic faces that normal form is now exact on the complete axial/polar fixed-node-norm tangent. For N occupied nodes of total complex internal dimension D, its unquotiented inertia is $(4D-2, 4D-2, 2D-N+2)$, and after node-phase reduction it is $(4D-2, 4D-2, 2D-2N+2)$. Every axisymmetric point also lies on an explicit nonaxisymmetric fixed-occupation arc. The transverse part is therefore hyperbolic on every automatic-face support stratum; what remains locally is to resolve the displayed radical into nonlinear semialgebraic components. On the active candidate-(16) component the separate restricted-current gate is now closed: both resonant nodes are q_-, so the complete axial/polar restriction is negative Kähler on every complex smooth stratum, with generic real rank 20. Its singular projective locus consists of two endpoint $\mathbb{CP}^4$ strata, and its complete fixed-occupation rotation-zero fibre is connected by Proposition 11.31; occupation gluing over every nonzero normalized scalar occupation is closed by Corollary 11.32. The cone origin remains excluded. On the candidate-(17)/(20) axisymmetric sections, exact occupation inequalities make the complete Zariski-tangent current nondegenerate, with affine inertia $(6, 10, 0)$ and projective inertia $(5, 9, 0)$; those section points remain algebraically singular. Away from that section, each active variety also contains an exact smooth bounded point with a surviving projective current radical. Hence neither is a global symplectic orbifold. Candidate (18) has the same global verdict by a different exact mechanism: its two rank-one parity channels have current eigenlines $(1, 1)$ and $(1, -1)$, the scalar cone crosses equal active occupation on the positive mixture $R_3 + s_{18}R_1$, and each resulting smooth bounded family has a four-complex-dimensional projective radical. The complete smooth affine presymplectic loci are classified by Proposition 11.16. For candidates (17) and (20), the complete complex singular carrier is also classified by Proposition 11.17: its two parity-product components are controlled by a common-square incidence resolution. Its real fixed-occupation intersection and lifted-rotation reduction remain open. The two algebraic singular-component images already meet physically at every positive occupation by Corollary 11.18; unlike candidate (18), their labels cannot imply quotient separation. The complete double-singular intersection is a connected rotation-zero hub at every positive occupation by Proposition 11.19. It remains open whether every component of either larger singular image meets this hub. One endpoint of that hub has the exact frequency-weighted bifurcation of Proposition 11.20: candidate (17) always reduces to one orbit, whereas candidate (20) acquires an interval quotient on its balance divisor. In particular, the unweighted inequality $N_- > N_+$ is not sufficient to exclude rotational balance. The complete candidate-(20) singular union on that balance divisor now contracts to the connected hub by Proposition 11.21, including the receiving-square vertex. Off balance, the same radial path covers the phase-real common-square sublocus, but its exact residual is not a nonradial no-go; complete singular connectedness remains open there and on candidate (17). Proposition 11.22 then classifies the complete uniform-scaling ansatz with a moving square direction. Candidate (17)'s $\alpha \leq 0$ stratum and candidate (20)'s off-balance $\alpha\delta > 0$ or $\alpha = 0$ strata contract to the hub; the opposite-sign non-phase-real region has an exact coefficient-zero obstruction within this ansatz. Proposition 11.23 classifies the enlarged fixed-direction independent-node ansatz. A path exists exactly when the coefficient-zero line meets the kernel-moment-zero locus inside the physical scaling square; positive-collinear incidence points contract by an explicit three-stage path, while generic fixed directions remain obstructed. Allowing the K-node directions themselves to deform gives the invariant normal form of Proposition 11.24: a point contracts exactly when its component of the compact admissible orbit space meets the coefficient-zero/kernel-moment-zero incidence. Both strict sign chambers contain explicit one-zero-node incidence points, but it remains open whether every admissible component reaches them. The regular fixed-occupation node-phase reductions are classified by Proposition 11.29. The latter retains the common parity-node phases and candidate-(18) spectators rather than quotienting them factorwise. On every smooth constant-corank fixed-occupation stratum, the lifted rotational moment map is basic along the current radical and the two reductions commute locally by Proposition 11.30. What remains is the global lifted-rotation zero-fibre geometry, global/Hausdorff leaf-space topology, real singular-locus reduction, constant-rank gluing and occupation gluing. Candidate (18)'s complete complex singular carrier is already resolved by Proposition 11.25, with all ten spectators retained; its real fixed-occupation current remains open. Corollary 11.26 shows that positive norms and all three lifted rotational constraints do not avoid any of these three singular carriers. A complete global theorem must therefore reduce the singular strata rather than restrict to the regular atlas. Within candidate (18)'s singular quotient, the two labelled components remain separated at every $N_- > 0$, giving the two-component lower bound of Proposition 11.27. The full zero fibre may still be connected through smooth strata; indeed, Proposition 11.28 joins one certified point in each singular component through the smooth central-m carrier at every positive occupation. It does not prove global connectedness. The four candidate-(19) and two candidate-(21) real linear sheets are closed separately: every core has current inertia $(5, 5, 0)$ and one connected fixed-occupation rotation-zero link, with spectator factors included. No distinct sheets are identified. The exact fibre-product equation is already certified; complex resonance components and occupation-level sums need not be recomputed. The symbolic-ℓ standard-branch resonance and collision theorem is now closed, as is the complete 164-coefficient two-momentum branch-basis census. The next exact enlargement is to classify the real connected components of the fixed-norm resonance strata after imposing all three lifted rotation moment maps. Occupation-level pairwise sums and complex resonance ideals need not be retested. This is followed by joins of distinct $|k|$ fibres without identifying their phase carriers or circumference backgrounds. The immediate physical target is to compare its strata with the transferred Einstein/extra/Maxwell interaction tensor and with observer response. A later causal theorem would replace the finite secular right inverses by support-controlled Green operators or prove that this cannot be done.

The intended final synthesis is not that every fourth-order mode disappears. It is that each mode receives an exact disposition:

$$\boxed{\text{linear mode} \;\to\; \text{moment map} \;\to\; \text{resonance} \;\to\; \text{interaction and observation.}}$$

17 Two-jet derived charge carrier

The full compact-Cauchy replacement for a fixed-group moment-map model is a constraint algebroid: the lapse bracket retains the inverse-metric structure function. The associated two-jet Kuranishi carrier is the Koszul derived zero fibre of the five charges, with $d\eta_A = \mu_A$. Its finite-harmonic restriction reproduces the five Taub maps in this paper.

This derived fibre is not compatible with the ambient linear Einstein/additional cofiber projection. The certified balanced Einstein-minus/extra tangent has total moment map zero while its two primary projections have nonzero opposite Hamiltonian charges. Hence the common zero fibre is intrinsically mixed; it is not the direct sum, nor the quotient, of separately neutral primary fibres. A later residual sequence must be formulated as a derived correspondence or fibre product. This statement is quadratic and does not remove the higher Kuranishi terms left open here.

18 Mixed-charge derived correspondence

The obstruction above has a minimal two-jet repair which does not reinstate a quotient of separately neutral branches. Let $E \to W \to X$ denote the certified pre-residual Einstein, Weyl and additional-primary sequence and let

$$\kappa_E : E \longrightarrow \mathfrak{o}, \qquad \kappa_X : X \longrightarrow \mathfrak{o}, \qquad \mathfrak{o} = \mathrm{span}\{H, P_x, J_1, J_2, J_3\}.$$

The correct branch-readable carrier is the homotopy pullback

$$\mathfrak{C} = (E \times X) \underset{\mathfrak{o}\times\mathfrak{o}}{\overset{h}{\times}} \mathfrak{o}_{\mathrm{anti}}, \qquad \Delta_{\mathrm{anti}}(c) = (c, -c).$$

Its strict Koszul presentation has coordinates $(e, x, c; \eta_E, \eta_X)$ and differential

$$d\eta_E = \kappa_E(e) - c, \qquad d\eta_X = \kappa_X(x) + c, \qquad d(e, x, c) = 0.$$

Thus $d^2 = 0$ exactly at the declared two-jet order. Moreover, with

$$\alpha = \eta_E + \eta_X, \quad \beta = \frac{1}{2}(\eta_X - \eta_E), \quad c' = c + \frac{1}{2}(\kappa_X - \kappa_E),$$

one has $d\alpha = \kappa_E + \kappa_X$ and $d\beta = c'$. The pair (β, c') is contractible, so this strict charge-transfer presentation is quasi-isomorphic to the five-generator Koszul model for the total moment map. It is nevertheless the useful presentation because c records the charge carried from one primary to the other.

On the real balanced two-amplitude fixture the strict tangent complex has dimensions $7 \to 10$, rank five, and hence

$$\dim H^0 = 2, \qquad \dim H^1 = 5.$$

The certified point is represented by

$$c = (\tau_E, 0, 0, 0, 0), \qquad \tau_E = \frac{48}{5}(-6 + 5\sqrt{3}), \qquad \tau_X = -\tau_E.$$

It maps to the total Weyl derived zero fibre, while its Einstein and extra projections have the nonzero charges c and $-c$. This simultaneously proves that the correspondence contains the balanced ray and that no map $\mathfrak{C} \to Z_E \times Z_X$ exists with the declared projections.

The map/form ledger is correspondingly asymmetric. The maps $\mathfrak{C} \to Z_W$, $\mathfrak{C} \to E$, and $\mathfrak{C} \to X$ are honest derived or ambient maps. The pullbacks $j^*\Omega_W$ and $p_X^* S_X$ are defined, where S_X is the lift-invariant Schur-complement form. But S_X is not obtained by quotienting the nonlinear correspondence, and a raw lifted extra Gram matrix remains lift dependent. The diagonal compact stabilizer action does descend: the transfer coordinate transforms coadjointly and the anti-diagonal equations are equivariant. No all-orders Kuranishi, bounded-evolution, causal, particle, positivity or quantum statement is inferred.

19 Third-order Kuranishi input gate

The two-jet correspondence fixes the correct carrier but does not by itself define its first higher obstruction. With the convention

$$\Phi(\epsilon) = \bar{\Phi} + \epsilon u + \epsilon^2 v + \epsilon^3 w + O(\epsilon^4),$$

the next equation and its projected constraint representative are

$$Lw = -D^2E[u, v] - \frac{1}{6}D^3E[u, u, u], \qquad K_3(u; v) = P_{\mathfrak{o}}\left(D^2C[u, v] + \frac{1}{6}D^3C[u, u, u]\right).$$

If v is replaced by $v + z$ with $Lz = 0$, then

$$K_3(u; v + z) - K_3(u; v) = P_{\mathfrak{o}} D^2C[u, z] = \ell_2(u, z).$$

The correction-independent object is therefore the class

$$[K_3(u; v)] \in \mathfrak{o} / \operatorname{im} \ell_2(u, -).$$

The earlier two-amplitude preflight saw only the Hamiltonian column of this map and hence rank one. On the complete same-frequency spin-two correction space, the $m = \pm 1$ directions add the J_1 and J_2 columns. The exact spin-two matrices give

$$\operatorname{im} \ell_2(u, -) = \operatorname{span}\{H, J_1, J_2\}, \qquad \operatorname{rank} \ell_2(u, -) = 3,$$

while P_x and J_3 vanish by $k = m = 0$. Thus the intrinsic global quotient has basis $\{P_x, J_3\}$ and dimension two.

The smallest finite third-order closure can be fixed before its coefficients are known. Its axial angular outputs are $\ell = 2, 4, 6$. Writing ω_- for the Einstein-minus frequency and ω_e for the additional frequency, its sixteen signed frequencies are

$$n_-\omega_- + n_e\omega_e, \qquad |n_-| + |n_e| \le 3, \qquad 3 - |n_-| - |n_e| \in 2\mathbb{Z}.$$

An exact shell comparison shows that the only coincidences are the original $\ell = 2$ shells: $(n_-, n_e) = (\pm 1, 0)$ on the Einstein-minus branch and $(0, \pm 1)$ on the additional branch. Thus no new kinematic shell type appears. The action-normalized coefficients can now also be evaluated.

Theorem 19.1 (Correction-independent global cubic class versus a representative-scoped bounded shell obstruction)**.** *Let u be the real balanced Einstein-minus/additional-primary tangent and let v be its certified no-homogeneous-addition second-order correction from the companion construction. Then:*

1. *the compact stabilizer representative vanishes componentwise,*

$$P_{\mathfrak{o}}\left(D^2C[u,v] + \frac{1}{6}D^3C[u,u,u]\right) = 0,$$

and hence $[K_3(u;v)] = 0$ in $\mathfrak{o}/\operatorname{im}\ell_2(u,-)$;

2. *for this certified v, in the bounded or finite-quasiperiodic correction class, each occupied original branch has a nonzero adjoint-shell functional. Numerically, in the declared action-row normalization, the positive-frequency values are*

$$R_-(u;v) = -2323.7892958977675\ldots,$$
$$(R_{e,1}(u;v), R_{e,2}(u;v)) = (0, -723.39760902920997\ldots).$$

The negative-frequency values are their reality conjugates. All nonzero claims are exact: the certificate records their algebraic expressions and minimal polynomials with nonzero constant coefficient;

3. *a finite exponential-polynomial third-order correction exists if secular terms are allowed. The bounded obstruction is therefore not a global compact-stabilizer Kuranishi obstruction.*

Proof. The selected action $\int \sqrt{-g}(3C^2/8 - F^2/4)$ supplies exact action-derived $q_2 = D^2E$ and $q_3 = D^3E$. Substitution of all 27 signed $\ell = 0, 2, 4$ correction rows gives the mixed term $D^2E[u,v]$; the four signed first-order modes give $D^3E[u,u,u]/6$. Their axial output has only even $\ell = 2, 4, 6$. Resolving its $\ell = 2$ dual-vector density requires the equatorial derivatives of orders $1, 3, 5$; the scalar rows use orders $0, 2, 4$. The fifth derivative is retained from the content-addressed order-ten checkpoint rather than inferred from the order-four public PBW payload. Deleting it changes the certified source and is a rejecting mutation.

The global projections vanish before any numerical simplification: H and P_x are scalar $\ell = 0$ covectors, whereas the lifted rotations are axial $\ell = 1$ covectors, so all pairings with the even-ℓ axial source vanish on the closed slice. The standard spin-two matrices then give the displayed rank-three image of $\ell_2(u,-)$. The action-derived arity-three Q^2 identity has zero defect on all four relevant Euler rows; a nonlinear gauge change therefore changes the representative only by the same ℓ_2 image and a closed-slice exact term.

Projection onto the four original shells gives the exact nonzero functionals recorded in `bridge/certificates/EINSTEIN_WEYL_COMPACT_CAUCHY_THIRD_ORDER_KURANISHI_EVALUATION_V1.json`.

Thus the Fredholm alternative forbids a bounded quasiperiodic primitive. For the finite exponential-polynomial class, the square axial constant-coefficient pencil has nonzero determinant proportional to $p^2 q$. Its adjugate reduces the system to scalar constant-coefficient equations, which are surjective on finite exponential-polynomials after polynomial factors in t are allowed. The required secular degree is at most one on the q shell and at most two on the p shell. This proves the three distinct assertions. Notice that only the quotient class in item 1 has been divided by the complete homogeneous second-order-correction freedom. Items 2 and 3 retain the explicitly declared representative v. $\square$

The theorem hierarchy is therefore:

Carrier and order	Correction class	Frozen disposition
Arbitrary $u \in V_{\rm fin}$, second order	finite exponential-polynomial	iff the five Taub moment maps vanish
Arbitrary $u \in V_{\rm fin}$, second order	bounded finite quasiperiodic	iff the complete finite $\{\mu, \mathcal{P}, \mathcal{R}\}$ ledger vanishes; its common real zero set is only partly classified
Balanced u, third-order global projection	quotient by all homogeneous second-order corrections	$[K_3] = 0$ in the two-dimensional quotient with basis $\{P_x, J_3\}$
Balanced u with certified v, third order	bounded finite quasiperiodic	obstructed on all four occupied original shells
Balanced u with certified v, third order	finite exponential-polynomial	solvable with secular degree at most one on q and at most two on p
Balanced u, arbitrary admissible second-order correction	bounded finite quasiperiodic	open

The theorem is deliberately not an all-orders result. It says that the quadratic charge balance survives the global cubic quotient, while uniform boundedness fails for the chosen second-order representative because of a different, local-in-time resonant cokernel. Causal/retarded inversion and the effect of allowing arbitrary homogeneous second-order additions on the shell representatives remain separate gates.

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